CYBER-PHYSICAL PROTOCOL TRAINING

ICS / OT / IoT Cyber-Physical Protocols

Field, control, supervisory, power, building and network protocols — how each one works, and how to assess it read-only.

Multiple slides per protocol: how it works, then its objects, functions and read-only enumeration.

Read-only by default · intrusive tests only in an isolated/representative environment with written authorization

ISA/IEC 62443 · SL-T = SL 3

SCOPE

Six protocol families

Grouped by where each protocol lives in the stack. Generic IT management protocols (SSH, SNMP, TLS…) are collected in the appendix.

Industrial automation & control

Modbus, EtherNet/IP & CIP, PROFINET/-BUS, S7, SRTP/EGD, UMAS, ANSL, K-sequence, POWERLINK, functional-safety comms

Supervisory & IIoT

OPC UA, OPC DA/DCOM, MQTT/Sparkplug, vendor SCADA middleware

Power & substation

DNP3, IEC 60870-5-104, IEC 61850 MMS/GOOSE/SV, IEC 62351, metering (ION/DLMS)

Building automation

BACnet/IP & MS/TP, BACnet/SC, LonWorks, Metasys N2, Siemens P2/FLN

Field, process & machine

HART, CAN / SAE J1939, vendor genset serial, legacy RS-485 (SY/MAX)

Network infrastructure

Industrial wireless, media redundancy, time sync, L2 discovery/provisioning

HOW TO READ THIS DECK

Multiple slides per protocol

Intro slide	Family, a plain-language summary, how it works (message model + frame/PDU), transport & ports, the native security posture, and example products that use it.
Data slides	The objects, registers, function codes, services or classes on the wire — how the protocol does read, write, command, poll, publish and subscribe.
Native read-only enumeration	The actual native requests to enumerate a device read-only — identity reads, status polls, model browses — plus the MITRE ATT&CK for ICS techniques in play.
Access tags	Each function is tagged R (read), W (write), Cmd (command), Pub/Sub (publish/subscribe) or Disc (discovery) so the read-only subset is obvious.

SAFETY RULE FOR EVERY ASSESSMENT

Work on live/production systems is **read-only**: native read views, **show** commands, exports and passive captures, plus native read/identity requests. Any write, scan, fuzz, brute-force or DoS runs **only** in an isolated or representative environment with written authorization. A mis-set drive can start a motor; a malformed frame can fault a controller.

INDUSTRIAL AUTOMATION & CONTROL

Modbus RTU / ASCII

The most widely deployed serial fieldbus protocol: a simple master/slave request-response over RS-485/232, with no session, no authentication and no encryption.

HOW IT WORKS

TRANSACTION MODEL

A single master polls each slave by its 1-byte address (1-247). The slave whose address matches replies; all others stay silent. One outstanding request at a time.

FRAME / PDU

RTU frame = **addr | function code | data | CRC-16**, delimited by 3.5-character idle gaps. ASCII wraps the same PDU in printable hex between **:** and CR/LF with an LRC.

MITRE ATT&CK for ICS **T0855 Unauthorized Command Message** · **T0836 Modify Parameter** · **T0868 Detect Operating Mode**

TRANSPORT & PORTS

RS-485 (2/4-wire) or RS-232

Typ. 9600-115200 8-N-1

Addr 1-247, broadcast 0

CRC-16 (RTU) / LRC (ASCII)

SECURITY POSTURE

- No authentication - any node can issue writes
- No encryption or integrity beyond the CRC
- Address is identity; trivially spoofed
- Control = physical bus access + device parameter/keypad lock

EXAMPLE PRODUCTS

ABB ACS/ACH/ACQ580 VFD

AutomationDirect DL06

Productivity2000 P2-550

Basler DGC-2020HD / MTU MGC-3000

Benshaw RSi H2 VFD

Cummins PowerCommand PCC3300

INDUSTRIAL AUTOMATION & CONTROL · DATA & OPERATIONS

Modbus RTU / ASCII — data model & operations

Four data tables, a handful of function codes, and a register map - that is the whole protocol. Learn the tables and the codes and you can read any Modbus device.

FUNCTIONS & SERVICES

Operation	Code / where	Access
Read Coils	FC 01	R
Read Discrete Inputs	FC 02	R
Read Holding Registers	FC 03	R
Read Input Registers	FC 04	R
Write Single Coil	FC 05	W
Write Single Register	FC 06	W
Write Multiple Registers	FC 16 (0x10)	W
Report Slave ID	FC 17 (0x11)	R
Read Device ID	FC 43/14 (0x2B/0x0E)	R

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

Coils (0xxxx) — 1-bit, read/write - digital outputs / control bits

Discrete Inputs (1xxxx) — 1-bit, read-only - digital inputs / status

Input Registers (3xxxx) — 16-bit, read-only - measured values

Holding Registers (4xxxx) — 16-bit, read/write - setpoints, parameters, control words

HOW READ / WRITE / COMMAND / POLL WORKS

Read: Master sends addr + FC03 + start register + quantity; slave replies byte-count + register values. One request, one response.

Write / command: FC06 sets one register, FC16 a block (a drive run/stop is often a bit in a control-register write). The slave echoes the request as acknowledgement.

Poll loop: The master repeats reads on a cycle - there is no publish/subscribe and no unsolicited data.

INDUSTRIAL AUTOMATION & CONTROL · ON THE WIRE

Modbus RTU / ASCII — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
Address	1 byte	Slave address 1-247 (0 = broadcast)
Function code	1 byte	FC01-FC43 selects the operation
Data	n bytes	Start address, quantity, or register/coil values
CRC-16	2 bytes	Error check, low byte first

EXAMPLE — READ HOLDING REGISTERS (FC 03) — 2 REGISTERS FROM SLAVE 1

-> 01 03 00 00 00 02 C4 0B

slave 01 | FC 03 | start 0x0000 | qty 0x0002 | CRC

<- 01 03 04 00 0A 00 14 7A 92

slave 01 | FC 03 | bytes 04 | reg0=0x000A(10) | reg1=0x0014(20) | CRC

NATIVE READ-ONLY ENUMERATION

Read device identity: FC 43 / 0x2B/0x0E (vendor, product, revision)

Report Slave ID: FC 17 (0x11)

Map live ranges with FC03 / FC04 reads from an authorized master

Passive capture at an RS-485 tap decodes addr + FC + data

Write codes (FC05/06/16) - observe only; never exercise on a live bus

INDUSTRIAL AUTOMATION & CONTROL

Modbus TCP

Modbus PDU carried over TCP port 502. The serial CRC is dropped (TCP provides integrity) and a 7-byte MBAP header is prepended - but the security model is unchanged: none.

HOW IT WORKS

TRANSACTION MODEL

A client opens a TCP session to the server on 502 and sends requests; the server answers. Multiple clients and outstanding transactions are allowed, matched by the MBAP transaction id.

FRAME / PDU

MBAP (txn id | proto id | length | unit id) | function code | data. Same function codes as RTU. The unit id routes to a serial slave behind a gateway. No CRC - TCP handles delivery.

MITRE ATT&CK for ICS **T0855 Unauthorized Command Message** · **T0846 Remote System Discovery** · **T0885 Commonly Used Port** · **T0814 Denial of Service**

TRANSPORT & PORTS

TCP 502 (server listens)

MBAP header + Modbus PDU

Unit id routes through gateways

No TLS on standard Modbus/TCP

SECURITY POSTURE

- No authentication - any host reaching 502 can read and write
- Cleartext; no integrity beyond TCP checksums
- No session identity, no replay protection
- Control = segmentation, client allow-list, device lock

EXAMPLE PRODUCTS

ABB ACS/ACH/ACQ580 VFD

AutomationDirect DL06

DirectSOFT

Productivity2000 P2-550

Productivity Suite

B&R Automation Studio / PVI

Modbus TCP — data model & operations

Identical data model and function codes to RTU - the only change on the wire is the 7-byte MBAP header in place of the CRC, and TCP in place of the serial line.

FUNCTIONS & SERVICES

Operation	Code / where	Access
Read Holding Registers	FC 03	R
Read Input Registers	FC 04	R
Read Coils / Discrete In	FC 01 / 02	R
Write Single / Multiple	FC 06 / 16	W
Report Slave ID	FC 17 (0x11)	R
Read Device ID	FC 43/14 (0x2B/0x0E)	R

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

Same four tables — Coils, Discrete Inputs, Input Registers, Holding Registers - as RTU

MBAP header — Transaction id | Protocol id (0) | Length | Unit id - prepended to the PDU

Unit id — Routes a request to a serial slave behind a Modbus/TCP gateway

Register map — Vendor-specific; the device manual defines what each 4xxxx/3xxxx address means

HOW READ / WRITE / COMMAND / POLL WORKS

Connect: Client opens TCP to 502; each request carries a transaction id so several can be outstanding at once.

Read / write: Same PDUs as RTU inside the MBAP envelope; the server answers per transaction id.

Gateway hop: A request with unit id N is forwarded to serial slave N behind the gateway.

Modbus TCP — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
Transaction ID	2 bytes	Echoed by server to match the reply
Protocol ID	2 bytes	Always 0x0000 for Modbus
Length	2 bytes	Byte count of unit id + PDU that follow
Unit ID	1 byte	Routes to a serial slave behind a gateway
Function + data	n bytes	Same PDU as RTU (no CRC)

EXAMPLE — READ HOLDING REGISTERS (FC 03) OVER TCP 502

-> 00 01 00 00 00 06 01 03 00 00 00 02

txn 1 | proto 0 | len 6 | unit 1 | FC03 | start 0 | qty 2

<- 00 01 00 00 00 07 01 03 04 00 0A 00 14

txn 1 | len 7 | unit 1 | FC03 | bytes 4 | 10 | 20

NATIVE READ-ONLY ENUMERATION

TCP connect to 502, then FC 43 (0x2B/0x0E) read device identification

FC 17 (0x11) Report Slave ID

FC 03/04 register reads to map the live address space (read-only)

MBAP transaction-id matching confirms a live Modbus server

Passive capture decodes MBAP + FC + register values

INDUSTRIAL AUTOMATION & CONTROL

EtherNet/IP & CIP

ODVA's Common Industrial Protocol (CIP) carried over Ethernet. Explicit messaging on TCP 44818, real-time implicit I/O on UDP 2222, device discovery via ListIdentity.

HOW IT WORKS

OBJECT MODEL

CIP is object-oriented: devices expose classes, instances and attributes addressed by class/instance/attribute IDs. Services (Get/Set Attribute, Forward_Open, MSG) act on them.

EXPLICIT VS IMPLICIT

Explicit = connected/unconnected request-response over TCP 44818 (config, programming, MSG). Implicit = cyclic class-1 I/O over UDP 2222, set up by a Forward_Open. ListIdentity returns vendor/product/serial/rev.

MITRE ATT&CK for ICS **T0846 Remote System Discovery** · **T0861 Point & Tag Identification** · **T0855 Unauthorized Command Message** · **T0831 Manipulation of Control**

TRANSPORT & PORTS

TCP 44818 - explicit messaging

UDP 2222 - class-1 I/O

UDP 44818 - ListIdentity discovery

CIP Security wraps these in TLS/DTLS

SECURITY POSTURE

- Base CIP has no authentication or encryption
- Identity from ListIdentity is unauthenticated
- Forward_Open / MSG accepted without credentials
- Control = CIP Security, keyswitch, segmentation

EXAMPLE PRODUCTS

ABB ACS/ACH/ACQ580 VFD

Productivity2000 P2-550

Productivity Suite

Hirschmann RSP25 (HiOS)

Benshaw RSi H2 VFD

CompactLogix 5380

EtherNet/IP & CIP — data model & operations

CIP is a catalogue of objects addressed by class / instance / attribute, reached with a small set of services. Discovery and reads need no credentials.

FUNCTIONS & SERVICES

Operation	Code / where	Access
ListIdentity	encap 0x63	Disc
ListServices / Interfaces	0x04 / 0x64	Disc
RegisterSession	encap 0x65	R
Get_Attributes_All	0x01	R
Get_Attribute_Single	0x0E	R
Set_Attribute_Single	0x10	W
Forward_Open (open I/O)	0x54	Cmd
Read Tag / Write Tag	0x4C / 0x4D	R/W

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

Identity object (0x01) — Vendor, product code, revision, serial, status - the device nameplate

TCP/IP (0xF5) & Ethernet Link (0xF6) — Network config and interface counters

Assembly (0x04) / Connection Mgr (0x06) — I/O data blocks and the Forward_Open that sets up cyclic I/O

Symbol / Template (0x6B / 0x6C) — Logix controller tag directory and tag structure

HOW READ / WRITE / COMMAND / POLL WORKS

Discover: ListIdentity (UDP 44818) returns the nameplate of every device that answers - no session needed.

Explicit read: RegisterSession on TCP 44818, then Get_Attributes_All on the Identity object, then read tags via the Symbol object.

Implicit I/O: A Forward_Open opens a class-1 connection; cyclic I/O then flows on UDP 2222.

INDUSTRIAL AUTOMATION & CONTROL · ON THE WIRE

EtherNet/IP & CIP — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
Command	2 bytes LE	e.g. 0x0063 ListIdentity, 0x0065 RegisterSession
Length	2 bytes LE	Bytes of the command-specific data
Session handle	4 bytes	0 for ListIdentity; assigned by RegisterSession
Status / context / options	16 bytes	Status, sender context, options
CPF items	n bytes	Identity object: vendor, product, revision, serial, name

EXAMPLE — LISTIDENTITY (UDP 44818) — UNAUTHENTICATED NAMEPLATE

-> 63 00 00 00 00 00 00 00 00 00 00 00 ...

command 0x0063 | length 0 | session 0

<- 63 00 <len> ... Identity item:

vendor id | device type | product code | revision | serial | 'product name'

NATIVE READ-ONLY ENUMERATION

ListIdentity (0x63) and ListServices (0x04) - unauthenticated nameplate + services

Get_Attributes_All (0x01) on Identity object (0x01)

Read TCP/IP (0xF5) and Ethernet Link (0xF6) objects for config

Symbol object (0x6B) to list controller tags (read-only)

Set_* / Write Tag (0x10 / 0x4D) are writes - do not issue on live

INDUSTRIAL AUTOMATION & CONTROL

CIP Security

The security extension for EtherNet/IP: TLS (explicit) and DTLS (implicit) with device certificates and integrity, managed through FactoryTalk Policy Manager zones and conduits.

HOW IT WORKS

WHAT IT ADDS

Authenticated, integrity-checked and optionally confidential CIP. Devices enroll certificates via FactoryTalk System Services (a CA); Policy Manager defines zones/conduits. Protection Mode makes a device refuse unauthenticated connections.

WHERE IT APPLIES

Controller firmware v31/33+ with Studio 5000 v32/33+. Not universal: many drives and older controllers have no CIP Security and fall back to segmentation. Enabling some features can void a device's certification level.

MITRE ATT&CK for ICS **T0830 Adversary-in-the-Middle** · **T0869 Standard Application Layer Protocol** · **T0822 External Remote Services**

TRANSPORT & PORTS

TCP/UDP 2221 - CIP Security (TLS/DTLS)

Certs enrolled via FT System Services

Policy Manager = zones + conduits

Ports ON by default on some CPUs (5590)

SECURITY POSTURE

- Adds device authentication + message integrity
- Optional confidentiality (TLS)
- Trust is only as good as the CA + enrollment
- Capability present != configured - verify it is enabled

EXAMPLE PRODUCTS

CompactLogix 5380 · ControlLogix 5580 · ControlLogix 5590 · PowerFlex 525 · PowerFlex 527 · PowerFlex 6000T MV

CIP Security — data model & operations

CIP Security does not change the object model - it wraps the same CIP in TLS/DTLS and adds security objects that hold the policy, certificates and Protection Mode state.

FUNCTIONS & SERVICES

Operation	Code / where	Access
TLS / DTLS handshake	port 2221	R
Get_Attribute_Single (sec objects)	0x0E	R
Read Protection Mode attr	0x0E	R
Policy / conduit (Policy Manager)	FT tool	R

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

CIP Security object (0x5D) — Overall security configuration / profile state

EtherNet/IP Security (0x5E) — TLS/DTLS settings for the transport

Certificate Management (0x5F) — Device certificate, CA chain, enrollment status

Protection Mode — When on, the device accepts only CIP-Security-authenticated connections

HOW READ / WRITE / COMMAND / POLL WORKS

Authenticate: Each endpoint presents its certificate; the peer validates it against the enrolled CA before any CIP flows.

Enforce: With Protection Mode on, unauthenticated CIP (tooling, scanners) is refused outright.

Manage: FactoryTalk Policy Manager defines which devices (zones) trust which, over which conduits.

INDUSTRIAL AUTOMATION & CONTROL · ON THE WIRE

CIP Security — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
TLS/DTLS record	5 bytes	Content type, version, length on port 2221
Handshake	n bytes	ClientHello/ServerHello, Certificate, key exchange
Device certificate	n bytes	X.509 identity enrolled via FactoryTalk CA
Encrypted CIP	n bytes	Normal CIP, now inside the authenticated channel

EXAMPLE — CIP WRAPPED IN TLS ON 2221 (PROTECTION MODE ON)

-> TCP 2221 ClientHello (TLS 1.2)

<- ServerHello + Certificate (device cert, CA = FT System Services)

cipher negotiated; both sides validate certs

unauthenticated CIP to 44818 -> refused when Protection Mode is on

NATIVE READ-ONLY ENUMERATION

Confirm 2221 answers with TLS/DTLS, not cleartext CIP

Read the Security / Protection-Mode attributes (Get_Attribute_Single 0x0E)

Inspect the device certificate presented in the TLS handshake (validity, key size, CA)

Review zones/conduits in FactoryTalk Policy Manager

Record any conduit still carrying unauthenticated CIP

INDUSTRIAL AUTOMATION & CONTROL

PROFINET

PROFIBUS & PROFINET International's Ethernet protocol: layered real-time classes from TCP/IP config down to isochronous I/O, with DCP for discovery and addressing.

HOW IT WORKS

REAL-TIME CLASSES

NRT (TCP/IP) for parameterization; RT (class 1/2) is cyclic Layer-2 traffic (EtherType 0x8892) bypassing IP for determinism; IRT (class 3) is hardware-scheduled isochronous. DCP (L2) assigns names/IP and can blink/reset devices.

ADDRESSING

Devices are addressed by a station name, not just IP. A tool or controller resolves the name via DCP at startup. GSDML files describe each device's modules and parameters.

MITRE ATT&CK for ICS **T0842 Network Sniffing** · **T0846 Remote System Discovery** · **T0856 Spoof Reporting Message** · **T0814 Denial of Service**

TRANSPORT & PORTS

RT: L2 EtherType 0x8892 (no IP)

DCP: L2 discovery / naming

NRT: TCP/UDP (config, OPC UA, web)

PN-CM 34964; SNMP for topology

SECURITY POSTURE

- RT/IRT have no authentication or encryption
- DCP identify/set is unauthenticated
- PN Security is emerging, rarely deployed
- Control = segmentation, managed switches, physical access

EXAMPLE PRODUCTS

ABB ACS/ACH/ACQ580 VFD

Hirschmann RSP25 (HiOS)

Danfoss VLT FC102

Emerson PACSystems RX3i

E+H Promag P300/P500

Phoenix Contact FL NAT 2000

PROFINET — data model & operations

PROFINET addresses devices by name, not just IP, and DCP is how names and addresses are discovered and set - all in cleartext at Layer 2.

FUNCTIONS & SERVICES

Operation	Code / where	Access
DCP Identify All	FrameID 0xFEFE, ServiceID 5	Disc
DCP Get (read params)	ServiceID 3	R
DCP Set (name / IP)	ServiceID 4	Cmd
Cyclic RT I/O	EtherType 0x8892	Pub/Sub
Alarm / acyclic	NRT	R

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

Station name — The real identity; a controller resolves it to a device via DCP at startup

DCP options/suboptions — IP params (0x01), Device Properties (0x02), DHCP, Control - read or set over L2

GSDML modules — Slot/subslot I/O layout and parameters, described per device type

Cyclic RT frame — EtherType 0x8892, a FrameID per connection carrying I/O each cycle

HOW READ / WRITE / COMMAND / POLL WORKS

Discover: A DCP Identify-All request (FrameID 0xFEFE, option 0xFF/0xFF) makes every station answer with name, IP and role.

Start up: The controller matches names to devices, (re)assigns IP via DCP Set, then begins cyclic RT exchange.

Run: Each cycle the controller and device publish their I/O frames on the Layer-2 segment.

INDUSTRIAL AUTOMATION & CONTROL · ON THE WIRE

PROFINET — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
Dst / Src MAC	12 bytes	Layer-2; DCP is not routed
EtherType	2 bytes	0x8892 (PROFINET RT / DCP)
FrameID	2 bytes	0xFEFE = DCP Identify multicast
Service ID / Type	2 bytes	ServiceID 5 = Identify, Type 0 = request
Blocks	n bytes	Option/suboption 0xFF/0xFF = all

EXAMPLE — DCP IDENTIFY-ALL — ENUMERATE STATIONS (LAYER 2)

-> EtherType 0x8892 | FrameID 0xFEFE | ServiceID 5 | Type 0 | AllSelector FF FF

<- each station replies:

station name 'plc1' | IP 192.168.0.10 | vendor/device id | role

(cyclic RT I/O then runs on 0x8892 per connection)

NATIVE READ-ONLY ENUMERATION

DCP Identify-All (FrameID 0xFEFE, ServiceID 5) lists every station read-only

DCP Get (ServiceID 3) reads a station's IP/name params

Passive capture of cyclic RT (0x8892) maps the live exchange

LLDP / SNMP for neighbor topology

DCP Set (ServiceID 4) changes name/IP - never issue on a live segment

INDUSTRIAL AUTOMATION & CONTROL

PROFIBUS DP

The classic RS-485 fieldbus behind PROFINET: a deterministic token-passing / master-slave bus for cyclic I/O between a controller and field devices and drives.

HOW IT WORKS

BUS ACCESS

Hybrid token bus: active masters pass a token (multi-master), and the token-holder polls its slaves cyclically (DP). Each device has a bus address (0-126). GSD files describe device capabilities and I/O sizing.

DP SERVICES

DP-V0 cyclic I/O; DP-V1 adds acyclic read/write of parameters (used by drive/instrument DTMs); DP-V2 adds isochronous and slave-to-slave. PA is the intrinsically-safe process variant over MBP.

MITRE ATT&CK for ICS **T0855 Unauthorized Command Message** · **T0836 Modify Parameter** · **T0814 Denial of Service**

TRANSPORT & PORTS

RS-485 (DP) up to 12 Mbit/s

Address 0-126

Token bus + master/slave polling

MBP physical layer for PROFIBUS PA

SECURITY POSTURE

- No authentication or encryption
- Bus address is identity; physical access = control
- Acyclic DP-V1 can write parameters
- Control = cabinet/physical security + segmentation

EXAMPLE PRODUCTS

Danfoss VLT FC102

PowerFlex 6000T MV

PROFIBUS DP — data model & operations

A master parameterizes each slave from its GSD file, then exchanges I/O cyclically; DP-V1 adds acyclic parameter access used by drive and instrument DTMs.

FUNCTIONS & SERVICES

Operation	Code / where	Access
Set_Prm / Chk_Cfg	parameterize	Cmd
Data_Exchange (cyclic I/O)	DP-V0	Pub/Sub
DP-V1 Read	acyclic	R
DP-V1 Write	acyclic	W
Slave_Diag (diagnostics)	read	R

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

Bus address (0-126) — The slave's identity on the RS-485 segment

Input / output data — Cyclic process image, sized by the GSD configuration

DP-V1 acyclic objects — Slot/index-addressed parameters for DTM read/write

Diagnostics — Standard + device-specific diagnostic blocks the master can read

HOW READ / WRITE / COMMAND / POLL WORKS

Parameterize: At startup the master sends Set_Prm / Chk_Cfg to each slave per its GSD, then the slave enters data exchange.

Exchange: Cyclic Data_Exchange moves the I/O image every bus cycle; the token rotates among any active masters.

Acyclic: A DTM reads or writes a parameter via DP-V1 without disturbing the cyclic I/O.

INDUSTRIAL AUTOMATION & CONTROL · ON THE WIRE

PROFIBUS DP — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
SD2	1 byte	0x68 start delimiter (variable-length telegram)
LE / LEr	2 bytes	Length, repeated
DA / SA	2 bytes	Destination + source bus address (0-126)
FC	1 byte	Frame control (e.g. Data_Exchange)
Data unit	n bytes	I/O or DP-V1 read/write payload
FCS / ED	2 bytes	Checksum + 0x16 end delimiter

EXAMPLE — DP CYCLIC DATA_EXCHANGE WITH SLAVE 3

-> 68 LE LEr 68 | DA=03 SA=02 | FC | output image | FCS 16

<- 68 LE LEr 68 | DA=02 SA=03 | FC | input image | FCS 16

DP-V1 Read (acyclic): slot/index addressed parameter -> value

NATIVE READ-ONLY ENUMERATION

Observe cyclic Data_Exchange and read Slave_Diag via the master (read-only)

Inventory bus addresses and GSD-declared modules

DP-V1 Read of parameters via the DTM (read-only)

Capture with a PROFIBUS tester/tap on the segment

Set_Prm / DP-V1 Write change the device - isolated bench only

INDUSTRIAL AUTOMATION & CONTROL

Siemens S7comm / S7comm-Plus

Siemens' proprietary control protocol to SIMATIC S7 PLCs over ISO-on-TCP (RFC 1006), used by TIA Portal and HMIs to read/write memory, blocks and diagnostics.

HOW IT WORKS

TRANSPORT & PDU

S7comm rides ISO-on-TCP: TCP 102 carries TPKT + COTP, then the S7 PDU (job/ack/userdata). Functions read/write data blocks, start/stop the CPU, upload/download blocks and read the diagnostic buffer.

S7COMM-PLUS

S7-1200/1500 use S7comm-Plus, which adds an anti-replay / integrity handshake that blocked naive replay. Access-protection levels and password-protected / know-how-protected blocks are set in TIA Portal; OPC UA and PN security are firmware-dependent.

MITRE ATT&CK for ICS **T0846 Remote System Discovery** · **T0868 Detect Operating Mode** · **T0831 Manipulation of Control** · **T0816 Device Restart/Shutdown**

TRANSPORT & PORTS

TCP 102 - ISO-on-TCP (TPKT/COTP)

S7comm (300/400) / S7comm-Plus (1200/1500)

OPC UA 4840, web 80/443, SNMP

PROFINET for I/O

SECURITY POSTURE

- S7comm (legacy) has no real authentication
- S7comm-Plus adds integrity/anti-replay, not confidentiality by default
- Access levels + block passwords are the native control
- F-CPU safety and Secure Comm need current firmware

EXAMPLE PRODUCTS

Siemens S7-1200 / S7-1500

INDUSTRIAL AUTOMATION & CONTROL · DATA & OPERATIONS

Siemens S7comm / S7comm-Plus — data model & operations

S7comm reads and writes CPU memory areas and system-status lists over a COTP/TPKT session; S7comm-Plus adds an integrity token but exposes the same data areas.

FUNCTIONS & SERVICES

Operation	Code / where	Access
Setup Communication	negotiate PDU	R
Read Var	0x04	R
Write Var	0x05	W
Read SZL (status)	SZL 0x001C / 0x0011 / 0x0074	R
PLC Stop / Start	0x29 / 0x28	Cmd
Upload / Download block	-	R/W

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

Memory areas — Inputs (I/E), Outputs (Q/A), Flags (M), Data Blocks (DB), Timers, Counters

Data Blocks (DB) — The structured program data read/written by Read Var / Write Var

SZL system-status lists — Component id, module id, CPU features and status - the device inventory

Protection level — Access-protection setting + know-how / block passwords set in TIA Portal

HOW READ / WRITE / COMMAND / POLL WORKS

Connect: TPKT/COTP connection request on TCP 102 carries the TSAP pair that selects the CPU (rack/slot).

Setup: An S7 Setup-Communication negotiates PDU size; then Read Var / Read SZL pull data.

Plus: On S7-1200/1500 (S7comm-Plus) each request carries an integrity token that defeats naive replay.

INDUSTRIAL AUTOMATION & CONTROL · ON THE WIRE

Siemens S7comm / S7comm-Plus — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
TPKT	4 bytes	03 00 + total length (ISO-on-TCP)
COTP	3+ bytes	Connection / data transport; CR carries the TSAP
S7 header	10-12 bytes	ROSCTR (job/ack), PDU ref, param/data length
Parameter	n bytes	Function (Read Var 0x04, Read SZL), item spec
Data	n bytes	Returned values / SZL records

EXAMPLE — COTP CONNECT (TSAP SELECTS THE CPU) THEN READ SZL

-> 03 00 00 16 11 E0 00 00 00 01 00 C0 01 0A C1 02 01 00 C2 02 01 02

TPKT | COTP CR | src TSAP 0x0100 | dst TSAP 0x0102 (S7-300/400)

(S7-1200/1500 use dst TSAP 0x0100 and S7comm-Plus)

-> Read SZL 0x0011 (module id) / 0x001C (component id) <- CPU identity records

NATIVE READ-ONLY ENUMERATION

COTP connect with the TSAP pair (0x0100->0x0102 S7-300/400; ->0x0100 S7-1200/1500)

S7 Setup Communication to negotiate the session

Read SZL 0x001C (component id), 0x0011 (module id), 0x0074 (CPU status)

Read Var (0x04) of specific DB/memory (read-only)

Stop/Start (0x29/0x28) and Download are control/write - never on live

INDUSTRIAL AUTOMATION & CONTROL

Emerson SRTP & EGD

The native protocols of Emerson (GE) PACSystems: SRTP service requests over TCP, and EGD (Ethernet Global Data) producer/consumer exchange - both cleartext.

HOW IT WORKS

SRTP

Service Request Transfer Protocol: a client/server request-response for reading and writing controller memory (%R, %I, %Q...), program and config, used by PAC Machine Edition. Cleartext, gated only by the controller privilege level + password.

EGD

Ethernet Global Data: a producer/consumer model where a controller periodically sends an exchange of tag data to subscribers - efficient many-to-many status, with no authentication.

MITRE ATT&CK for ICS **T0861 Point & Tag Identification** · **T0855 Unauthorized Command Message** · **T0842 Network Sniffing**

TRANSPORT & PORTS

SRTP: TCP 18245 / 18246

EGD: UDP 18246 (producer/consumer)

Modbus/TCP + OPC UA also available

PROFINET on some CPUs

SECURITY POSTURE

- Neither SRTP nor EGD is authenticated or encrypted
- Privilege levels 1-4 + passwords gate SRTP writes
- EGD has no subscriber authentication
- Control = SDG hardening, segmentation, TPM/signed firmware where present

EXAMPLE PRODUCTS

Emerson PACSystems RX3i

Emerson SRTP & EGD — data model & operations

SRTP is request-response reads/writes of PACSystems reference memory; EGD is a periodic producer/consumer push of tag exchanges. Both name memory by %-reference.

FUNCTIONS & SERVICES

Operation	Code / where	Access
SRTP Init	type 0x00	Disc
Read PLC Info	type 0x02	R
Read memory	SRTP read	R
Write memory	SRTP write	W
EGD produce	periodic	Pub
EGD consume	subscribe	Sub

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

Reference memory — %R registers, %I/%Q discrete I/O, %AI/%AQ analog, %M internal - the PACSystems address space

SRTP service requests — Typed requests: init, read PLC info, read/write memory

EGD exchange — A producer id + exchange id + a fixed list of tags, sent on a period

Privilege levels 1-4 — Gate SRTP write/control; reads are typically open

HOW READ / WRITE / COMMAND / POLL WORKS

SRTP: Client connects (18245/18246), sends Init, then Read PLC Info and memory reads; writes need the privilege level.

EGD: A controller produces an exchange of tags on a period to one or more consumers - no request, no acknowledgement.

Mix: A cell often runs SRTP for engineering and EGD for fast many-to-many status at once.

INDUSTRIAL AUTOMATION & CONTROL · ON THE WIRE

Emerson SRTP & EGD — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
SRTP message	56 bytes	Fixed service-request header over TCP
Service request type	1 byte	0x00 init, 0x02 read PLC info, read/write memory
Memory reference	n bytes	%R / %I / %Q / %AI segment + offset + length
EGD exchange	n bytes	Producer id + exchange id + fixed tag list (UDP push)

EXAMPLE — SRTP INIT + READ PLC INFO (TCP 18245)

-> SRTP Init (type 0x00) <- session established

-> Read PLC Info (type 0x02) <- CPU model, firmware, run state

-> Read %R100 length 2 <- register values

EGD: producer 0x0A010203 pushes exchange 1 {%AI tags} on a period

NATIVE READ-ONLY ENUMERATION

SRTP Init (type 0x00) then Read PLC Info (type 0x02) - read-only identity

SRTP memory reads of %R/%AI from an authorized client

Passively map EGD exchanges (producer id + exchange id + tag list)

Ports 18245/18246 (SRTP), UDP 18246 (EGD)

SRTP writes need privilege - do not exercise on live

INDUSTRIAL AUTOMATION & CONTROL

Schneider UMAS (over Modbus)

The proprietary engineering protocol of Schneider Modicon controllers, tunneled inside Modbus function code 90 - the channel Control Expert uses to program and control M340/M580.

HOW IT WORKS

HOW IT RIDES MODBUS

UMAS requests are carried as Modbus/TCP FC90 (0x5A) payloads on port 502. Beyond register reads, UMAS handles project download/upload, start/stop and memory access - the real control plane behind the Modbus facade.

SECURITY MODES

M580 adds Standard / Secured / Enforced cybersecurity modes and an application password; Enforced makes the CPU HTTPS-only and refuses 502/UMAS. ModiPwn (CVE-2021-22779) abused UMAS session handling - the application password is the primary mitigation.

MITRE ATT&CK for ICS **T0843 Program Download** · **T0831 Manipulation of Control** · **T0855 Unauthorized Command Message**

TRANSPORT & PORTS

TCP 502 - Modbus carrying UMAS FC90

HTTPS 443 on M580 (Enforced = HTTPS-only)

FTP/TFTP/SNMP on some CPUs

IPsec via BMENUA0100 (M580)

SECURITY POSTURE

- Legacy UMAS is unauthenticated (pre-fix)
- Application password is the key native control
- Enforced mode refuses cleartext 502/UMAS
- Patch ModiPwn / Modbus auth-bypass CVEs

EXAMPLE PRODUCTS

Studio 5000

Modicon M580

EcoStruxure Control Expert

Modicon M340

INDUSTRIAL AUTOMATION & CONTROL · DATA & OPERATIONS

Schneider UMAS (over Modbus) — data model & operations

UMAS is a function set tunneled inside Modbus FC90 (0x5A). Standard Modbus reads still work; the engineering operations live in the FC90 sub-function byte.

FUNCTIONS & SERVICES

Operation	Code / where	Access
Read ID	UMAS 0x02	R
Read project / PLC info	UMAS 0x03 / 0x04	R
Read memory block	UMAS 0x20	R
Start / Stop PLC	UMAS 0x40 / 0x41	Cmd
Write memory	UMAS 0x22	W
Standard Modbus read	FC 03	R

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

Modbus FC90 (0x5A) — The carrier - every UMAS request is an FC90 payload on port 502

UMAS sub-function — A byte selecting the operation (read id, read project, read memory, start/stop, write)

Session / reservation — Secured controllers require a reservation + handshake before writes

Cybersecurity mode — Standard / Secured / Enforced - Enforced refuses 502 and forces HTTPS

HOW READ / WRITE / COMMAND / POLL WORKS

Carry: Control Expert sends a Modbus/TCP frame with FC90; the sub-function byte decides read vs. control.

Reserve: On a Secured CPU a reservation + application password gates project download and writes.

Enforce: In Enforced mode the CPU drops 502/UMAS entirely and engineering runs over HTTPS.

INDUSTRIAL AUTOMATION & CONTROL · ON THE WIRE

Schneider UMAS (over Modbus) — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
MBAP	7 bytes	Standard Modbus/TCP header, unit id
Function code	1 byte	0x5A (90) = UMAS carrier
UMAS sub-function	1 byte	0x02 ReadID, 0x03/04 project, 0x20 read memory
Session / payload	n bytes	Reservation handle (Secured) + request data

EXAMPLE — UMAS READ ID OVER MODBUS FC90 (TCP 502)

-> 00 01 00 00 00 06 00 5A 00 02

MBAP | FC 0x5A (90) | UMAS 0x02 (Read ID)

<- CPU model, firmware, project name, protection state

Enforced cybersecurity mode -> 502/UMAS refused; engineering over HTTPS

NATIVE READ-ONLY ENUMERATION

Standard Modbus FC03 register reads work regardless of UMAS

Observe FC90 ReadID / ReadProjectInfo in capture (read-only)

Confirm cybersecurity mode and that the application password is set

Verify FTP/TFTP disabled and the IP allow-list applied

UMAS sub-codes vary by firmware - confirm against Schneider behaviour; do not replay writes

⚠ UMAS sub-function codes are vendor-internal — confirm exact values against the installed firmware.

INDUSTRIAL AUTOMATION & CONTROL

B&R ANSL / INA

The engineering and runtime protocols of B&R (ABB) Automation Runtime: INA and its successor ANSL carry Automation Studio programming and PVI data to X20 / APC controllers.

HOW IT WORKS

INA -> ANSL

INA (11159) is the older transport; ANSL (11169) is current, with an optional TLS mode (11179). Automation Studio, PVI and the Runtime Utility Center use these to browse, download and monitor controllers.

DEFAULTS THAT MATTER

By default ANSL is configured without TLS and without user/password on the controller. Boot AR (recovery) allows unauthenticated INA/ANSL and open FTP - physical access to the reset button is a control. Record the AR version; defaults differ across AR 4.x vs 6.x.

MITRE ATT&CK for ICS **T0822 External Remote Services** · **T0843 Program Download** · **T0859 Valid Accounts**

TRANSPORT & PORTS

ANSL TCP 11169 / 11179 (TLS)

INA TCP 11159 (default/boot AR)

OPC UA 4840, SNMP, Modbus 502

Debugger 2323; VNC 5900+

SECURITY POSTURE

- ANSL not TLS-enabled by default; no user/pw by default
- Boot AR allows unauthenticated connect + open FTP
- Improper cert-validation CVEs in the AS client (patch to 6.5)
- Control = enable TLS + AR user roles, physical reset access

EXAMPLE PRODUCTS

B&R APROL PCS

B&R APC/PPC 3100

B&R Automation Studio / PVI

B&R X20 PLC

B&R mapp View HMI

INDUSTRIAL AUTOMATION & CONTROL · DATA & OPERATIONS

B&R ANSL / INA — data model & operations

ANSL/INA carry Automation Studio's browse-and-read of the controller's process-variable namespace. The framing is proprietary, so enumeration is service-level, not packet-level.

FUNCTIONS & SERVICES

Operation	Code / where	Access
Browse targets	PVI / AS	Disc
Read process variable	PVI read	R
Read module list	AS online	R
Download / write	AS	W
Boot AR connect	INA 11159	R

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

Process variables (PV) — The named AR variables browsed and read by PVI / Automation Studio

Module / task list — The installed software objects on the controller

AR user roles — The user/role system - off by default unless configured

TLS config — ANSL over TLS on 11179 when enabled; plain 11169 otherwise

HOW READ / WRITE / COMMAND / POLL WORKS

Connect: Automation Studio / PVI opens ANSL (11169) or legacy INA (11159) to the controller; TLS on 11179 if configured.

Browse & read: The tool browses the PV namespace and reads values and the module/task list.

Boot AR: In recovery, INA/ANSL accept unauthenticated connections and FTP is open - a physical-access concern.

INDUSTRIAL AUTOMATION & CONTROL · ON THE WIRE

B&R ANSL / INA — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
TCP transport	-	ANSL 11169 (TLS 11179) / INA 11159
Session	-	Optional AR user/password + TLS (off by default)
Request	-	Browse / read process variable / read module list
Payload	-	Proprietary — not publicly dissected

EXAMPLE — PVI / AUTOMATION STUDIO BROWSE-AND-READ (SERVICE LEVEL)

-> TCP connect 11169 (or 11159 INA / boot AR)

-> browse process-variable namespace <- PV list

-> read PV 'gMotor.speed' <- value

on the Linux host: ss -tulpn lists the live services

NATIVE READ-ONLY ENUMERATION

TCP connect 11169 / 11159 to confirm the service is live

Browse and read process variables via PVI (read-only)

On the Linux host: ss -tulpn to list the actual services

Confirm ANSL TLS + AR user roles are enabled and Boot AR is unreachable

No public ANSL packet dissector - confirm framing against B&R docs

⚠ ANSL/INA framing is proprietary — enumerate at the service level and confirm against B&R documentation.

INDUSTRIAL AUTOMATION & CONTROL

Koyo K-sequence / DirectNET / ECOM

The native protocols of AutomationDirect DirectLOGIC PLCs: K-sequence and DirectNET over serial, plus the Host Engineering ECOM protocol over Ethernet (UDP 28784).

HOW IT WORKS

SERIAL PROTOCOLS

K-sequence is the native programming/monitoring protocol on the RS-232 port (used by DirectSOFT); DirectNET is a simple master/slave data protocol on the second serial port. Both are cleartext with only the PLC's multilevel program password as a control.

ECOM OVER ETHERNET

The H0/H2/H4-ECOM modules speak the proprietary Host Engineering protocol on UDP 28784 (0x7070, fixed). ECOM100 adds Modbus/TCP (client & server) on 502. NetEdit configures modules over UDP. No TLS anywhere.

MITRE ATT&CK for ICS **T0855 Unauthorized Command Message** · **T0868 Detect Operating Mode** · **T0836 Modify Parameter**

TRANSPORT & PORTS

Serial: K-sequence, DirectNET (RS-232/422/485)

ECOM: UDP 28784 (0x7070, fixed)

ECOM100 also Modbus/TCP 502

NetEdit over UDP

SECURITY POSTURE

- No user accounts, no encryption
- DirectSOFT multilevel program password is the only native lock
- ECOM protocol + NetEdit are unauthenticated
- Control = segmentation + physical mode switch (RUN/TERM/STOP)

EXAMPLE PRODUCTS

AutomationDirect DL06 | DirectSOFT

Koyo K-sequence / DirectNET / ECOM — data model & operations

DirectLOGIC memory is octal-addressed (V-memory, X/Y I/O, C relays). K-sequence and DirectNET read it over serial; ECOM100 also answers standard Modbus/TCP.

FUNCTIONS & SERVICES

Operation	Code / where	Access
K-sequence read/write	serial	R/W
DirectNET read/write	serial	R/W
ECOM read	UDP 28784	R
ECOM100 Modbus read	FC 03 / 04	R
ECOM100 Modbus write	FC 06 / 16	W

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

V-memory (V, octal) — 16-bit data words - the main register space

X / Y (octal) — Discrete inputs (X) and outputs (Y)

C control relays — Internal bit storage

ECOM map — ECOM100 exposes V-memory/I-O through a Modbus register mapping

HOW READ / WRITE / COMMAND / POLL WORKS

Serial: DirectSOFT over K-sequence reads/writes V-memory and I/O directly on the RS-232 port.

ECOM: On an ECOM module the Host Engineering protocol answers on UDP 28784; ECOM100 adds Modbus/TCP on 502.

Poll: A SCADA master polls V-memory via Modbus on the ECOM100 register map.

Koyo K-sequence / DirectNET / ECOM — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
ECOM (UDP 28784)	-	Host Engineering protocol, port 0x7070 fixed
K-sequence / DirectNET	-	Serial read/write of V-memory (octal)
ECOM100 Modbus	-	V-memory exposed via a Modbus register map on 502
V-memory	16-bit words	Octal-addressed data; X/Y I/O; C relays

EXAMPLE — ECOM100 MODBUS READ OF V-MEMORY (TCP 502)

-> 00 01 00 00 00 06 01 03 00 00 00 02 (FC03 read 2 words)

<- 00 01 00 00 00 07 01 03 04 00 0A 00 14

NetEdit (UDP 28784) read-only -> module IP/firmware/identity

confirm the V-memory -> Modbus register mapping for the model

NATIVE READ-ONLY ENUMERATION

ECOM100: Modbus FC03/FC04 reads of the V-memory mapping

FC 43 (0x2B/0x0E) device identification on the ECOM100

NetEdit read-only module info over UDP 28784

Confirm the program password is set and the mode switch position

Writes (K-seq write, FC06/16) - observe only on live

INDUSTRIAL AUTOMATION & CONTROL

POWERLINK, openSAFETY & X2X

The B&R real-time stack: Ethernet POWERLINK for deterministic I/O, openSAFETY for functional-safety data over it, and X2X Link for local backplane/remote I/O.

HOW IT WORKS

POWERLINK

A deterministic Ethernet protocol: a managing node polls controlled nodes in isochronous cycles (slotted access) on standard Ethernet hardware. Addressed by node ID; an asynchronous phase carries TCP/IP/ASnd for config.

OPENSAFETY & X2X

openSAFETY is a bus-independent black-channel safety protocol (SafetyDomains, SN addresses, CRC + time-stamp) that rides POWERLINK over UDP 9877. X2X Link is B&R's local I/O bus between CPU and modules.

MITRE ATT&CK for ICS **T0842 Network Sniffing** · **T0814 Denial of Service** · **T0880 Loss of Safety**

TRANSPORT & PORTS

POWERLINK: L2 real-time (managing node)

openSAFETY over UDP 9877

X2X / X2X+ local I/O bus

ASnd / TCP-IP in async phase

SECURITY POSTURE

- POWERLINK has no authentication/encryption
- openSAFETY protects safety integrity (CRC+timestamp), not security
- Node ID is identity; physical/segment control applies
- Safety data must fail to a safe state independent of the network

EXAMPLE PRODUCTS

B&R APC/PPC 3100

B&R X20 PLC

POWERLINK, openSAFETY & X2X — data model & operations

A managing node polls every controlled node once per isochronous cycle (PReq/PRes); an async phase carries SDO reads of the object dictionary and IP traffic.

FUNCTIONS & SERVICES

Operation	Code / where	Access
SoC / SoA (cycle)	managing node	Cmd
PReq / PRes (I/O)	cyclic	Pub/Sub
SDO read	ASnd / UDP	R
SDO write	ASnd / UDP	W
Ident response	ASnd	Disc

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

Node ID — MN = 240, CNs = 1-239 - the cyclic-slot identity

PDO (process data) — The cyclic I/O image exchanged each PReq/PRes

Object dictionary / SDO — Indexed objects read/written acyclically via ASnd/UDP

openSAFETY SPDO — Safety payload with its own SN address, CRC and timestamp

HOW READ / WRITE / COMMAND / POLL WORKS

Cycle: The MN sends Start-of-Cycle, then polls each CN (PReq) which answers (PRes) in its slot - deterministic timing.

Async: After the cyclic phase, SDO and IP traffic run; SDO reads/writes the object dictionary.

Safety: openSAFETY SPDOs ride the same wire; the consumer fails safe on a bad CRC/timestamp.

POWERLINK, openSAFETY & X2X — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
Dst / Src MAC	12 bytes	Layer-2 real-time
EtherType	2 bytes	0x88AB (Ethernet POWERLINK)
Message type	1 byte	SoC / PReq / PRes / SoA / ASnd
Node ID	1 byte	MN 240, CN 1-239
PDO / SDO	n bytes	Cyclic process data / acyclic object-dictionary access

EXAMPLE — ONE ISOCHRONOUS CYCLE + AN SDO READ

MN -> SoC (start of cycle)

MN -> PReq node 1 CN1 -> PRes node 1 (its I/O) ... each CN in turn

MN -> SoA (async slot)

-> SDO read object 0x1000 <- object value (openSAFETY SPDO rides the same wire)

NATIVE READ-ONLY ENUMERATION

Passive capture of the cyclic PReq/PRes maps node IDs and I/O

SDO read of object-dictionary indices (read-only)

Ident responses identify node roles

openSAFETY SafetyDomains: record, never alter

No frame injection on a live POWERLINK segment

INDUSTRIAL AUTOMATION & CONTROL

Functional-safety comms: CIP Safety & PROFIsafe

Safety protocols layered on standard networks (the black channel) to carry SIL-rated data between safety controllers, drives and I/O - with integrity, not confidentiality.

HOW IT WORKS

BLACK-CHANNEL MODEL

CIP Safety (over EtherNet/IP) and PROFIsafe (over PROFINET/PROFIBUS) add a safety layer - sequence number, timestamp/watchdog, safety CRC and a connection ID - so a corrupted or delayed frame is detected and the consumer goes to a safe state. The base network is untrusted by design.

WHY IT MATTERS

These protocols guarantee **safety integrity**, not security: they do not authenticate an engineering change. A GuardLogix safety signature, PROFIsafe F-parameters and safety locks are the artifacts to record. Firmware changes on safety controllers require impact analysis + a regenerated signature.

MITRE ATT&CK for ICS **T0880 Loss of Safety** · **T0858 Change Operating Mode** · **T0838 Modify Alarm Settings**

TRANSPORT & PORTS

CIP Safety over EtherNet/IP (2222 I/O)

PROFIsafe over PROFINET/PROFIBUS

openSAFETY over POWERLINK/UDP 9877

Safety CRC + seq + timestamp + conn ID

SECURITY POSTURE

- Integrity & timeliness assured; no confidentiality
- Does not authenticate engineering changes
- Safety signature / F-parameters define the trusted config
- Bypasses, forces, Battle Short = safety-significant findings

EXAMPLE PRODUCTS

ABB ACS/ACH/ACQ580 VFD

CompactLogix 5380

ControlLogix 5580

ControlLogix 5590

ControlLogix 5570

Studio 5000

INDUSTRIAL AUTOMATION & CONTROL · DATA & OPERATIONS

Functional-safety comms: CIP Safety & PROFIsafe — data model & operations

Safety comms do not add new read/write verbs - they add a wrapper (seq, timestamp, safety CRC, connection id) so the consumer detects tampering and fails safe. What you enumerate is the safety configuration.

FUNCTIONS & SERVICES

Operation	Code / where	Access
Safety producer/consumer	cyclic	Pub/Sub
Read safety signature	engineering tool	R
Read F-parameters	engineering tool	R
Read safety-lock state	engineering tool	R

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

Safety connection — A producer/consumer pair bound by a connection id / SCID

Safety PDU wrapper — Sequence number + timestamp/watchdog + second CRC over the safety data

PROFIsafe F-parameters — F_Dest_Add, F_WD_Time and friends - the device's safety address and watchdog

Safety signature — GuardLogix signature ID + date; the fingerprint of the trusted safety config

HOW READ / WRITE / COMMAND / POLL WORKS

Carry: The safety data rides the base network (EtherNet/IP, PROFINET) inside its safety wrapper - the network is untrusted.

Validate: The consumer checks sequence, timestamp and safety CRC each cycle; any fault drives outputs to the safe state.

Change control: A firmware or logic change requires impact analysis and a regenerated safety signature.

INDUSTRIAL AUTOMATION & CONTROL · ON THE WIRE

Functional-safety comms: CIP Safety & PROFIsafe — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
Base network frame	-	CIP Safety on EtherNet/IP, PROFIsafe on PROFINET
Safety data	n bytes	The SIL-rated process/output data
Sequence / time stamp	n bytes	Detects lost, old or out-of-order frames
Connection ID / F-addr	n bytes	PROFIsafe F_Dest_Add; CIP Safety connection
Safety CRC	2-4 bytes	Second CRC over the safety PDU

EXAMPLE — SAFETY PRODUCER/CONSUMER WITH A SAFETY SIGNATURE

producer -> safety data + seq + timestamp + safety CRC (each cycle)

consumer validates seq/timestamp/CRC -> else SAFE STATE

read-only: GuardLogix Safety Signature = ID 16#A3F2 , date 2026-05-01

PROFIsafe F-parameters: F_Dest_Add=7 , F_WD_Time=150 ms (record, do not alter)

NATIVE READ-ONLY ENUMERATION

Read the safety signature ID + date and the safety-lock state (read-only); store the signature separately separately

Record PROFIsafe F-parameters without altering them

Capture the base CIP/PROFINET as usual; the safety wrapper rides it

Treat forces, bypasses and Battle Short as safety-significant findings

Never suggest an action needing unlock or re-signing

SUPERVISORY & IIOT

OPC UA

The modern, vendor-neutral interoperability standard: a service-oriented information model with a real, built-in security model - when it is turned on.

HOW IT WORKS

INFORMATION MODEL & SERVICES

Servers expose an address space of nodes (objects, variables, methods) that clients browse, read, write, call and subscribe to. Transport is binary opc.tcp (4840) or HTTPS. Discovery via a Local Discovery Server / GetEndpoints.

SECURITY MODEL

Per-endpoint SecurityPolicy (None / Basic256Sha256 / Aes256...) with signing and/or encryption, X.509 application certificates for mutual trust, and user authentication (anonymous / username / certificate). The risk is misconfiguration: Anonymous + SecurityPolicy None.

MITRE ATT&CK for ICS **T0861 Point & Tag Identification** · **T0830 Adversary-in-the-Middle** · **T0811 Data from Information Repositories**

TRANSPORT & PORTS

opc.tcp TCP 4840 (binary) or HTTPS

Kepware default opc.tcp 49320; Ignition 62541

SecurityPolicy per endpoint

PubSub over MQTT/UDP (optional)

SECURITY POSTURE

- Real security model: sign/encrypt + certs + user auth
- But often deployed as Anonymous / None - verify
- Cert trust lists must be curated
- Strong policy = Basic256Sha256+, >=2048 RSA / ECC

EXAMPLE PRODUCTS

AVEVA System Platform

B&R APROL PCS

B&R APC/PPC 3100

B&R Automation Studio / PVI

B&R X20 PLC

B&R mapp View HMI

OPC UA — data model & operations

Every OPC UA server is a browsable graph of nodes reached by a fixed service set. The endpoint advertises its own security - so the first read is: what security does it require?

FUNCTIONS & SERVICES

Operation	Code / where	Access
Hello / OpenSecureChannel	HEL / OPN	Disc
GetEndpoints	discovery	Disc
Browse	view	R
Read	attribute	R
Write	attribute	W
Call	method	Cmd
CreateSubscription + MonitoredItem	-	Sub

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

AddressSpace nodes — Objects, Variables, Methods - the browsable graph, each with a NodeId + namespace

References — HasComponent, Organizes, HasTypeDefinition - how nodes relate

Endpoints — Each with a SecurityPolicy (None / Basic256Sha256 ...) and UserTokenPolicy

Subscriptions — MonitoredItems that push value changes to a client

HOW READ / WRITE / COMMAND / POLL WORKS

Handshake: opc.tcp Hello (HEL) negotiates buffers; OpenSecureChannel applies the endpoint's SecurityPolicy.

Session: CreateSession + ActivateSession carries the user token (anonymous / username / certificate).

Use: Browse the address space, Read/Write attributes, or Subscribe to MonitoredItems for change-driven data.

OPC UA — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
Message type	3 bytes	'HEL' Hello, 'OPN' open channel, 'MSG' service
Chunk / size	5 bytes	Chunk type + message length
Hello body	n bytes	Protocol ver, buffer sizes, max message, endpoint URL
SecurityPolicy	string	Per endpoint: None / Basic256Sha256 / Aes256...
UserTokenPolicy	n	Anonymous / UserName / Certificate

EXAMPLE — HELLO + GETENDPOINTS — READ THE SECURITY REQUIREMENT

-> HELF <size> ver=0 rcv/snd buf ... endpointUrl 'opc.tcp://plc:4840'

<- ACKF <size> (buffers agreed)

-> GetEndpoints <- endpoints + SecurityPolicy + UserTokenPolicy

SecurityPolicy=None + Anonymous = finding; browse/read only, no Write/Call

NATIVE READ-ONLY ENUMERATION

opc.tcp Hello (HEL), then GetEndpoints - lists SecurityPolicies + UserTokenPolicies

Confirm the endpoint's SecurityPolicy is not None and user auth is required

Anonymous Browse (expect denied if hardened); UaExpert read-only browse

Inspect the server certificate (validity, key size, trust)

Write / Call are control - do not issue on live nodes

SUPERVISORY & IIOT

OPC DA / Classic (DCOM)

The legacy OPC standard built on Microsoft COM/DCOM: still pervasive between SCADA, historians and connectivity servers - and a notorious firewall and hardening problem.

HOW IT WORKS

DCOM TRANSPORT

OPC DA marshals calls over DCOM: an RPC endpoint mapper on TCP 135 plus dynamically-negotiated high ports. Security is Windows authentication (NTLM/Kerberos) and DCOM launch/access permissions - powerful but brittle, and often loosened to make it work.

WHY IT PERSISTS

Connectivity servers (Kepware, FactoryTalk, PSO/PME) still expose OPC DA for legacy clients. The 2021 DCOM hardening patch broke many OPC DA links, so sites sometimes weaken it - the opposite of what SL-3 wants.

MITRE ATT&CK for ICS **T0842 Network Sniffing** · **T0859 Valid Accounts** · **T0868 Detect Operating Mode**

TRANSPORT & PORTS

DCOM: TCP 135 (RPC EPM) + dynamic ports

Windows auth (NTLM/Kerberos)

DCOM launch/access ACLs

Tunnellers (OPC UA gateways) preferred

SECURITY POSTURE

- Security = Windows/DCOM ACLs; easy to over-permit
- Wide dynamic port range complicates segmentation
- Anonymous/loosened DCOM is a real finding
- Migrate to OPC UA or a tunneller where possible

EXAMPLE PRODUCTS

Kepware KEPServerEX | Red Lion N-View | FactoryTalk | Studio 5000 | EcoStruxure PowerSCADA | EcoStruxure PME

OPC DA / Classic (DCOM) — data model & operations

OPC DA is COM objects reached over DCOM: a server exposes groups of items (tags) you browse and read. The security is entirely Windows/DCOM, enumerated through the RPC endpoint mapper.

FUNCTIONS & SERVICES

Operation	Code / where	Access
GetStatus	IOPCServer	R
Browse address space	IOPCBrowse.. .	Disc
Read (sync/async)	IOPCSyncIO	R
Write	IOPCSyncIO	W
AddGroup / AddItems	-	R

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

OPCServer / OPCGroup / OPCItem — Server -> groups -> items (tags), the DA object hierarchy

ProgID / CLSID — How a server is registered and launched on the host

DCOM ACLs — Launch and Access permissions - the real access control

RPC endpoint mapper — TCP 135 maps interface UUIDs to the dynamic ports

HOW READ / WRITE / COMMAND / POLL WORKS

Bind: The client resolves the server via the RPC endpoint mapper on 135, then activates it over DCOM on a dynamic port.

Browse & read: It browses the server address space to find items, adds a group, and reads item values.

Auth: Every step is gated by Windows authentication and the server's DCOM launch/access ACLs.

SUPERVISORY & IIOT · ON THE WIRE

OPC DA / Classic (DCOM) — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
DCE/RPC bind	-	To the endpoint mapper on TCP 135
Interface UUID	16 bytes	e1af8308-5d1f-11c9-91a4-08002b14a0fa (EPM)
Dynamic port	-	EPM returns the OPC server's high port
DCOM auth	-	NTLM/Kerberos + launch/access ACLs
OPC calls	-	GetStatus, Browse, Read on the activated server

EXAMPLE — ENUMERATE OPC SERVERS VIA THE RPC ENDPOINT MAPPER

-> DCE/RPC Bind to EPM (135), interface e1af8308-...

<- endpoint map: OPC server CLSID -> dynamic port

-> activate server -> Browse address space -> Read item 'Tank1.Level'

review DCOM launch/access ACLs and the authentication level

NATIVE READ-ONLY ENUMERATION

DCE/RPC bind to the endpoint mapper (UUID e1af8308-5d1f-11c9-91a4-08002b14a0fa) on 135 - read-only

135 - read-only endpoint list

Enumerate registered OPC servers (ProgIDs/CLSIDs) on the host

Review DCOM launch/access ACLs and the authentication level

Confirm the 2021 DCOM hardening was not weakened

Writes go through the same interfaces - do not exercise on live

SUPERVISORY & IIOT

MQTT & Sparkplug B

The lightweight publish/subscribe protocol of IIoT: clients publish to topics on a broker and subscribe to others. Sparkplug B adds an OT-specific topic namespace and state model.

HOW IT WORKS

PUB/SUB MODEL

Clients connect to a broker (1883, or 8883 for TLS) and publish/subscribe to hierarchical topics with QoS 0/1/2. The broker decouples producers and consumers and is the trust anchor and chokepoint.

SPARKPLUG B

An OT profile: a defined topic namespace (**spBv1.0/...**), birth/death certificates for state awareness, and efficient binary (protobuf) payloads - making MQTT usable as a real-time OT transport with edge nodes and devices.

MITRE ATT&CK for ICS **T0842 Network Sniffing** · **T0868 Detect Operating Mode** · **T0855 Unauthorized Command Message**

TRANSPORT & PORTS

TCP 1883 (plain) / 8883 (TLS)

Broker-mediated pub/sub, QoS 0/1/2

Sparkplug B topic namespace + STATE

WebSocket variants on 8080/8443

SECURITY POSTURE

- MQTT itself: username/password + optional TLS
- Plain 1883 = cleartext credentials & payloads
- Broker ACLs control topic access
- Use 8883 (TLS) + per-client auth + topic ACLs

EXAMPLE PRODUCTS

Productivity2000 P2-550 | Productivity Suite | B&R APROL PCS | PowerLogic ION9000 | PowerLogic PM8000 | Kepware KEPServerEX

SUPERVISORY & IIOT · DATA & OPERATIONS

MQTT & Sparkplug B — data model & operations

Everything is a topic on a broker. You publish to a topic or subscribe to a filter; Sparkplug B imposes a defined topic tree and birth/death state so OT tooling can make sense of it.

FUNCTIONS & SERVICES

Operation	Code / where	Access
CONNECT	client->broker	R
SUBSCRIBE	topic filter	Sub
PUBLISH	topic	Pub
Sparkplug NBIRTH/DBIRTH	state	Pub
Sparkplug NCMD/DCMD	command	Cmd
$SYS read	broker stats	R

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

Topic tree — Hierarchical, /-separated; subscribe with + (one level) and # (multi-level) wildcards

QoS 0/1/2 — At-most-once / at-least-once / exactly-once delivery

Retained + will — Last retained value per topic; a will message on unexpected disconnect

Sparkplug B — spBv1.0/<group>/<msgtype>/<node>/<device>; NBIRTH/DBIRTH state + protobuf metrics

HOW READ / WRITE / COMMAND / POLL WORKS

Connect: The client CONNECTs to the broker (1883 plain / 8883 TLS) with optional credentials.

Sub / pub: It SUBSCRIBEs to topic filters and receives matching PUBLISHes; publishers push to topics - the broker fans out.

Sparkplug: Edge nodes announce NBIRTH, stream NDATA, and accept NCMD/DCMD commands within the namespace.

SUPERVISORY & IIOT · ON THE WIRE

MQTT & Sparkplug B — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
Fixed header	1 byte	Packet type + flags: 0x10 CONNECT, 0x82 SUBSCRIBE, 0x30 PUBLISH
Remaining length	1-4 bytes	Variable-length encoded body size
Variable header	n bytes	Packet id, topic, protocol name/level
Payload	n bytes	Topic filters (SUBSCRIBE) or message (PUBLISH)

EXAMPLE — CONNECT THEN SUBSCRIBE $SYS/# (READ-ONLY)

-> 10 .. 'MQTT' lvl 4 flags keepalive clientId <- 20 02 00 00 CONNACK ok

-> 82 .. packetId topic '$SYS/#' qos 0 <- 90 .. SUBACK

<- 30 .. topic '$SYS/broker/version' payload 'x.y'

8883 = TLS; 1883 = cleartext creds; never PUBLISH to control topics

NATIVE READ-ONLY ENUMERATION

CONNECT then SUBSCRIBE $SYS/# for broker stats (read-only, authorized)

SUBSCRIBE read-only to observed topic filters to learn the tree

Confirm TLS on 8883 vs cleartext 1883, and per-client credentials

Review broker topic ACLs - who may publish command topics

Never PUBLISH to control / NCMD / DCMD topics on live

SUPERVISORY & IIOT

Vendor SCADA middleware

Proprietary transports that tie a vendor's SCADA/HMI stack together: AVEVA SuiteLink, Rockwell FactoryTalk (RPC/DCOM), and Ignition's Gateway Network.

HOW IT WORKS

SUITELINK & FACTORYTALK

AVEVA SuiteLink (TCP 5413) carries real-time tag data with quality/timestamp between InTouch/System Platform nodes. FactoryTalk Services run over RPC/DCOM (135 + dynamic) as the security backbone (Directory, Linx, Diagnostics) for Studio 5000 / FT View.

IGNITION GATEWAY NETWORK

Ignition uses HTTP(S) (8088/8043) for the gateway/designer and a Gateway Network (TCP 8060) between gateways, plus OPC UA (62541). 8.3 adds roles/permissions and secrets management; permissions do **not** auto-migrate 8.1->8.3 - re-verify.

MITRE ATT&CK for ICS **T0811 Data from Information Repositories** · **T0859 Valid Accounts** · **T0830 Adversary-in-the-Middle**

TRANSPORT & PORTS

SuiteLink TCP 5413

FactoryTalk RPC/DCOM 135 + dynamic

Ignition GAN TCP 8060; web 8088/8043

OPC UA 62541; activation 27000/1443

SECURITY POSTURE

- SuiteLink: no strong native crypto - segment
- FactoryTalk: prefer Network Directory + real accounts
- Ignition: enforce TLS, roles, and GAN authentication
- Patch platform (PSIRT / vendor advisories)

EXAMPLE PRODUCTS

AVEVA System Platform

CompactLogix 5380

ControlLogix 5580

ControlLogix 5590

Ignition 8.1/8.3

FactoryTalk

SUPERVISORY & IIOT · DATA & OPERATIONS

Vendor SCADA middleware — data model & operations

These are glue transports between SCADA components - tag feeds and directories, not device control. Enumeration is about the account/trust model and the ports they listen on.

FUNCTIONS & SERVICES

Operation	Code / where	Access
SuiteLink read tag	TCP 5413	R
FactoryTalk directory read	RPC/DCOM 135	R
Ignition OPC UA read	62541	R
GAN peer	TCP 8060	R
Activation	27000/1443	R

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

SuiteLink items — Real-time tag + quality + timestamp between InTouch/System Platform nodes

FactoryTalk Directory — Users, groups and security policy; Linx shortcuts to controllers

Ignition tags + GAN — Tag providers, roles/permissions, and gateway-to-gateway peering

OPC UA endpoint — Ignition's southbound/northbound OPC UA on 62541

HOW READ / WRITE / COMMAND / POLL WORKS

Subscribe: A client subscribes to real-time tags (SuiteLink / FT Linx / Ignition) and receives value+quality updates.

Authorize: FactoryTalk authenticates against its Directory (prefer Network Directory + real accounts).

Federate: Ignition gateways peer over the Gateway Network; OPC UA carries device data in and out.

SUPERVISORY & IIOT · ON THE WIRE

Vendor SCADA middleware — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
SuiteLink	TCP 5413	Real-time tag + quality + timestamp feed
FactoryTalk	RPC/DCOM 135	Directory + Linx over DCE/RPC
Ignition GAN	TCP 8060	Gateway-to-gateway; OPC UA on 62541
Auth	-	FT directory accounts; Ignition roles/permissions

EXAMPLE — ENUMERATE THE ACCOUNT/TRUST MODEL (READ-ONLY)

-> DCE/RPC to 135 -> FactoryTalk services + directory type

-> GetEndpoints on Ignition OPC UA 62541 -> security policy

confirm TLS on web (8043) + authenticated GAN peering

re-verify Ignition permissions after an 8.1 -> 8.3 upgrade

NATIVE READ-ONLY ENUMERATION

Enumerate FactoryTalk services via the RPC endpoint mapper on 135 (read-only)

GetEndpoints on Ignition OPC UA (62541) to read its security

Review the directory/user model and role assignments

Confirm TLS on web (8043) and authenticated GAN peering

Re-verify Ignition permissions after an 8.1->8.3 upgrade

⚠ These transports are vendor-specific — verify ports and the account model against the installed product.

POWER & SUBSTATION

DNP3 (+ Secure Authentication)

The dominant SCADA telemetry protocol in North American electric/water utilities: an event-driven master/outstation protocol with report-by-exception and time-stamped data.

HOW IT WORKS

OBJECT / EVENT MODEL

Outstations organize data as objects grouped by type (binary/analog inputs & outputs, counters) with static and event variations. Masters poll (Class 0 static) and receive unsolicited events (Class 1/2/3). Control via select-before-operate (SBO).

SECURE AUTHENTICATION

DNP3-SA (IEEE 1815) adds challenge-response HMAC authentication of critical requests (e.g. operate) over the same channel - integrity/authentication, not encryption. Without SA, DNP3 is cleartext and control messages are unauthenticated.

MITRE ATT&CK for ICS **T0855 Unauthorized Command Message** · **T0856 Spoof Reporting Message** · **T0836 Modify Parameter** · **T0814 Denial of Service**

Note: Master = Controlling Station

TRANSPORT & PORTS

TCP/UDP 20000 (default insecure) TCP/UDP 19999 (default secure TLS (or DTLS)

Master/outstation, report-by-exception

SBO for control outputs

DNP3-SA = HMAC challenge-response

SECURITY POSTURE

- Base DNP3: no authentication or encryption
- Outstation address is identity; spoofable
- SA adds auth of critical ops (if deployed)
- TLS via IEC 62351-3 for confidentiality

EXAMPLE PRODUCTS

PowerLogic ION9000 · Juniper SRX340

PowerLogic PM8000 · PowerLogic ION7650

ProSoft PLX3x/PLX32 · Qualitrol 118 ITM

POWER & SUBSTATION · DATA & OPERATIONS

DNP3 (+ Secure Authentication) — data model & operations

Data is groups of typed points (binary, analog, counter) in static and event variations; reads pull a class, control uses select-before-operate on an output group.

FUNCTIONS & SERVICES

Operation	Code / where	Access
Read	FC 1	R
Write	FC 2	W
Select	FC 3	Cmd
Operate	FC 4	Cmd
Direct Operate	FC 5	Cmd
Enable/Disable Unsolicited	FC 20 / 21	Cmd
Cold/Warm Restart	FC 13 / 14	Cmd

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

Object groups — Binary Input g1, Binary Output/CROB g10/g12, Analog Input g30, Analog Output g40/g41, Counter g20

Static vs event variations — Static = current value; event = change, delivered by class

Classes 0 / 1 / 2 / 3 — Class 0 = all static data; 1/2/3 = event priorities (report-by-exception)

Point index — Each point addressed by group + variation + index

HOW READ / WRITE / COMMAND / POLL WORKS

Integrity :poll The controlling station reads all data in all classes (1, 2, 3 & 0) to retrieve all events followed by all static data to initialize or refresh its data image. The outstation returns all pending events followed by every static point.

Report-by-exception: The controlling station reads all events by Class (each class independently or all together), or may enable the outstation to transmit events spontaneously, known as unsolicited reporting.

Control (SBO): Select (FC3) a g12 CROB point, then Operate (FC4) the outstation arms then acts

Note: Master = Controlling Station

POWER & SUBSTATION · ON THE WIRE

DNP3 (+ Secure Authentication) — on the wire

How it looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
Start	2 bytes	0x05 0x64 data-link header
Length / control	2 bytes	Frame length + DIR/PRM control
Dest / Src	4 bytes	Outstation + master link addresses
CRC	2 bytes	Per-block CRC
App layer	n bytes	App control, function (01 READ), objects (group/var/qualifier)

EXAMPLE — Integrity Poll (Object Group 60) Classes 1, 2, 3 & 0

-> 05 64 <len> C4 <dest>: <src> <crc> (data-link header)

app: C0 01 3C 02 06 3C 03 06 3C 04 06 3C 01 06

app ctrl | FUNC 01 Read | Obj 0x3C (60) Var 2 (Class 1) Qualifier 6 (all) | Obj 60 Var 3 (Class 2) Qual 6 | Obj 60 Var 4 (Class 3) Qual 6 | Obj 60 Var 1 (Class 0) Qual 6

<- outstation returns all buffered events in Classes 1, 2 & 3 followed by all static data

NATIVE READ-ONLY ENUMERATION

Integrity poll - Read of all data from all Classes

Data Link header 0x05 0x64 always appears as the beginning of a data link frame

Read device attributes (object group 0) for device information

Confirm whether DNP3-SA (challenge-response) is in use; Determine if DNP3 security is in use

Select/Operate or Direct Operate on CROB or Analog output blocks - never on live plant

Note: Master = Controlling Station

POWER & SUBSTATION

IEC 60870-5-104

The European counterpart to DNP3: IEC 60870-5-101 telecontrol ASDUs carried over TCP/IP (2404), widely used in transmission/distribution SCADA outside North America.

HOW IT WORKS

ASDU MODEL

Data is exchanged as Application Service Data Units typed by identifier (single/double point, measured values, commands) with an information-object address and a cause of transmission. APCI frames (I/S/U) manage the TCP connection and sequence numbers.

COMMANDS

Control uses command ASDUs (single/double command, set-point) often with select-then-execute semantics. Like DNP3, the base protocol has no authentication; IEC 62351 provides the security layer.

MITRE ATT&CK for ICS **T0855 Unauthorized Command Message** · **T0856 Spoof Reporting Message** · **T0842 Network Sniffing**

Note: Master = Controlling Station

TRANSPORT & PORTS

TCP 2404

APCI I/S/U frames + ASDUs

IOA addressing, cause of transmission

Secured via IEC 62351

SECURITY POSTURE

- No native authentication or encryption
- Optional authentication extensions defined in IEC 60870-5-7
- Common-address / IOA is identity
- Command ASDUs accepted without auth
- Confidentiality Option = IEC 62351 TLS + segmentation

EXAMPLE PRODUCTS

EcoStruxure PowerSCADA

SIPROTEC 5 7SJ85

SIPROTEC 5 7UM85

SIPROTEC 5 7UT85

POWER & SUBSTATION · DATA & OPERATIONS

IEC 60870-5-104 — data model & operations

Telemetry and commands are typed ASDUs carried in I-frames; the link is opened with STARTDT and the whole point list is pulled with a General Interrogation.

FUNCTIONS & SERVICES

Operation	Code / where	Access
STARTDT / STOPDT / TESTFR	U-frame	Cmd
General Interrogation	C_IC_NA_1 (100)	R
Single / Double command	C_SC/DC (45/46)	Cmd
Set-point	C_SE (48-50)	Cmd
Spontaneous report	COT 3	Pub

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

Note: Master = Controlling Station

DATA MODEL / OBJECTS

ASDU type IDs — M_SP_NA_1=1 single point, M_ME_NC_1=13 float, C_SC_NA_1=45 single command, C_IC_NA_1=100 interrogation

Information Object Address (IOA) — Addresses each point within the common address

Cause of Transmission (COT) — Why the ASDU was sent - e.g. 6 = activation, 20 = interrogated by station

APCI frames I/S/U — Manage the TCP link and sequence numbers

HOW READ / WRITE / COMMAND / POLL WORKS

Open: Client connects to 2404 and sends STARTDT act; the outstation answers STARTDT con.

Interrogate: A General Interrogation (type 100, COT 6, QOI 20) makes the outstation report all current points.

Run: Changes arrive spontaneously (COT 3); commands use select-then-execute ASDUs.

POWER & SUBSTATION · ON THE WIRE

IEC 60870-5-104 — on the wire

How it looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
Start	1 byte	0x68
APDU length	1 byte	Bytes that follow
Control field	4 bytes	I/S/U format + send/receive sequence numbers
Type ID	1 byte	e.g. 100 = C_IC_NA_1 interrogation, 13 = measured float
COT / ASDU	n bytes	Cause of transmission + information objects (IOA + value)

EXAMPLE — STARTDT THEN GENERAL INTERROGATION

-> 68 04 07 00 00 00 (STARTDT act, U-frame)

<- 68 04 0B 00 00 00 (STARTDT con)

-> 68 0E <ctrl> | TypeID 100 | COT 6 (activation) | ... | QOI 0x14 (20, station)

<- outstation reports all points (type 1/13...), then spontaneous updates (COT 3)

NATIVE READ-ONLY ENUMERATION

STARTDT act U-frame: 0x68 0x04 0x07 0x00 0x00 0x00

General Interrogation: TypeID 100, COT 6, QOI 20 - reads all points

Decode ASDU type IDs, IOAs and COT from the capture

Confirm whether IEC 62351 TLS wraps the link

Command ASDUs (45/46/48-50) - never inject on live

Note: Master = Controlling Station

POWER & SUBSTATION

IEC 61850 MMS

The station-bus protocol of the modern digital substation: client/server access to a standardized data model of IEDs, carried over MMS (Manufacturing Message Specification) on TCP.

HOW IT WORKS

STANDARDIZED MODEL

IEC 61850 defines a self-describing model - logical devices -> logical nodes (e.g. **XCBR** breaker) -> data objects -> attributes. Clients browse and read/write this model, and control breakers via select-before-operate, over MMS.

TRANSPORT

MMS rides over TCP 102 (ISO-on-TCP, RFC 1006). Reporting (buffered/unbuffered) pushes data changes to clients. SCL (SCD/CID/ICD) files describe the whole substation - prime reconnaissance material, handle as sensitive.

MITRE ATT&CK for ICS **T0861 Point & Tag Identification** · **T0855 Unauthorized Command Message** · **T0800 Activate Firmware Update Mode**

TRANSPORT & PORTS

TCP 102 - MMS (ISO-on-TCP)

Self-describing LN/DO/DA model

SBO control of primary plant

SecureTLS variant (62351) TCP 3782

SECURITY POSTURE

- MMS base: no authentication/encryption
- Model browsing reveals the full substation
- SCL files are sensitive design data
- Control = IEC 62351 (TLS + auth), segmentation

EXAMPLE PRODUCTS

PowerLogic ION9000 | SEL-3505 RTAC | SEL-3555 RTAC | SEL-3350 RTAC | EcoStruxure PowerSCADA | Siemens DIGSI 5

POWER & SUBSTATION · DATA & OPERATIONS

IEC 61850 MMS — data model & operations

The IED is a self-describing tree (logical device -> logical node -> data object -> attribute). You browse the names, read the values, and watch reports - all defined by the SCL files.

FUNCTIONS & SERVICES

Operation	Code / where	Access
Initiate (association)	MMS Initiate	R
GetNameList (browse)	MMS	Disc
Read	MMS Read	R
Write	MMS Write	W
Oper / SBOw (control)	Control	Cmd
Report (BRCB/URCB)	report	Pub

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

Logical Node (LN) — XCBR breaker, MMXU measurements, PTOC overcurrent - standardized function blocks

Data Object / Attribute — e.g. XCBR1.Pos.stVal - the breaker position status value

Datasets + reports — BRCB/URCB push a dataset of changes to subscribed clients

SCL files (CID/ICD/SCD) — The full offline description of the IED and substation

HOW READ / WRITE / COMMAND / POLL WORKS

Associate: COTP on TCP 102, then an MMS Initiate opens the association to the IED.

Browse & read: GetNameList walks the LD/LN/DO/DA tree; Read pulls status and measured values.

Control: Breaker operations use select-before-operate (SBOw then Oper) on the control DO.

POWER & SUBSTATION · ON THE WIRE

IEC 61850 MMS — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
TPKT / COTP	7 bytes	ISO-on-TCP on 102 (same as S7 transport)
MMS Initiate	n bytes	Opens the association to the IED
Service	n bytes	GetNameList, Read, Write, control Oper/SBOw
Object name	string	LD/LN/DO/DA, e.g. RELAY/XCBR1.Pos.stVal
Value	n bytes	Status / measurement / quality / timestamp

EXAMPLE — INITIATE + GETNAMELIST + READ (STATION BUS 102)

-> TPKT/COTP connect -> MMS Initiate <- association up

-> GetNameList (domain) <- RELAY/XCBR1, RELAY/MMXU1, ...

-> Read RELAY/XCBR1.Pos.stVal <- 2 (closed), quality good, timestamp

SCL (CID/ICD) describes the whole model offline; no Oper/SBOw on live

NATIVE READ-ONLY ENUMERATION

COTP connect (102) + MMS Initiate to open the association

GetNameList to browse the full LD/LN/DO/DA model (read-only)

Read status/measurement attributes (e.g. MMXU)

Parse the SCL (CID/ICD/SCD) offline for the complete model

Oper / SBOw on breakers - never on live plant

POWER & SUBSTATION

IEC 61850 GOOSE & Sampled Values

The real-time Layer-2 multicast protocols of the substation: GOOSE for fast protection trips/interlocks and Sampled Values for digitized CT/VT waveforms - microsecond-critical, unroutable.

HOW IT WORKS

GOOSE

Generic Object Oriented Substation Event: publisher/subscriber L2 multicast (EtherType 0x88B8) carrying binary status (trips, interlocks, breaker positions) with a state/sequence number and a retransmission scheme for sub-4 ms delivery. No IP - it stays on the station LAN.

SAMPLED VALUES

SV (EtherType 0x88BA, IEC 61850-9-2 / -9-2LE) streams digitized instantaneous current/voltage samples from merging units to protection IEDs at 80/256 samples per cycle - the process bus. Timing (PTP) and integrity are safety-critical.

MITRE ATT&CK for ICS **T0842 Network Sniffing** · **T0856 Spoof Reporting Message** · **T0880 Loss of Safety** · **T0814 Denial of Service**

TRANSPORT & PORTS

GOOSE: L2 multicast EtherType 0x88B8

SV: L2 multicast EtherType 0x88BA

Process/station bus, no IP

PTP (1588) for sample timing

SECURITY POSTURE

- No authentication or encryption natively
- Spoofed GOOSE can command false trips
- 62351-6 adds authentication; rarely deployed
- Control = dedicated process-bus VLAN, physical access

EXAMPLE PRODUCTS

SIPROTEC 5 7SJ85 | SIPROTEC 5 7UM85 | SIPROTEC 5 7UT85

POWER & SUBSTATION · DATA & OPERATIONS

IEC 61850 GOOSE & Sampled Values — data model & operations

Both are publish-only Layer-2 multicast with no request side: GOOSE carries a dataset of status that re-fires fast on change; SV streams a fixed sample rate of current/voltage.

FUNCTIONS & SERVICES

Operation	Code / where	Access
GOOSE publish	multicast 0x88B8	Pub
GOOSE subscribe (IED config)	-	Sub
SV publish (merging unit)	multicast 0x88BA	Pub
SV subscribe (protection IED)	-	Sub

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

GOOSE control block (GoCB) — Names the dataset, GOOSE ID (gocbRef), AppID and VLAN

stNum / sqNum — State number (increments on change) and sequence number (retransmits)

SV stream — svID, smpCnt and the ASDU of current/voltage samples per cycle

EtherType — GOOSE 0x88B8, SV 0x88BA - Layer 2, no IP

HOW READ / WRITE / COMMAND / POLL WORKS

Steady state: A publisher re-sends its GOOSE at a slow heartbeat with a fixed stNum.

Event: On a status change stNum increments and the frame re-fires in a fast burst (ms) for reliable delivery.

SV: A merging unit streams 80 or 256 samples per cycle to protection IEDs on the process bus.

POWER & SUBSTATION · ON THE WIRE

IEC 61850 GOOSE & Sampled Values — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
Dst MAC	6 bytes	Multicast 01:0C:CD:01:xx:xx (GOOSE) / 01:0C:CD:04 (SV)
EtherType	2 bytes	0x88B8 GOOSE / 0x88BA SV
APPID / length	4 bytes	Application id + PDU length
GOOSE PDU	n bytes	gocbRef, stNum, sqNum, dataset values
SV ASDU	n bytes	svID, smpCnt, sample set (I/V)

EXAMPLE — GOOSE PUBLISH — STATE CHANGE RE-FIRES FAST

<- eth 0x88B8 | APPID 0x0001 | gocbRef 'RELAY/LLN0GOgcb01'

stNum 7 sqNum 0 (status just changed; sqNum climbs as it retransmits)

dataset: XCBR1.Pos.stVal = TRIP -> subscribers act within ms

SV: svID 'MU01' smpCnt 1200 | 4 samples/ASDU | 80 or 256 per cycle

NATIVE READ-ONLY ENUMERATION

Passive L2 capture only (eth.type 0x88B8 / 0x88BA)

Map GoCB / svID, AppID, VLAN and dataset members

Verify stNum/sqNum behaviour and publisher-subscriber pairs

Cross-reference publishers/subscribers against the SCL

Never publish GOOSE/SV frames on a live bus

POWER & SUBSTATION

IEC 62351 (security for 61850 / DNP3 / 104)

Not a data protocol but the security layer for the power-system protocols above: TLS, authentication and integrity added to DNP3, IEC 60870 and IEC 61850.

HOW IT WORKS

WHAT IT COVERS

62351-3 = TLS for TCP profiles (DNP3, 104, MMS). 62351-4 = security for MMS/application. 62351-5 = authentication for DNP3/101/104 (the basis of DNP3-SA). 62351-6 = authentication for GOOSE/SV. 62351-8 = role-based access control.

REALITY ON THE GROUND

Support is uneven and often off by default. SEL RTACs, SIPROTEC 5 and some meters implement SecureTLS/authentication; many field links still run cleartext. The assessment records which parts are actually enabled and where compensating segmentation carries the rest.

MITRE ATT&CK for ICS **T0830 Adversary-in-the-Middle** · **T0869 Standard Application Layer Protocol** · **T0859 Valid Accounts**

TRANSPORT & PORTS

62351-3 TLS (MMS 3782, DNP3/104)

62351-5 auth (DNP3-SA)

62351-6 GOOSE/SV authentication

62351-8 RBAC

SECURITY POSTURE

- Adds TLS + authentication + RBAC when enabled
- Frequently unsupported or disabled - verify
- SecureTLS ports (MMS 3782) indicate use
- Where absent -> compensating segmentation + VPN

EXAMPLE PRODUCTS

SEL-3505 RTAC | SEL-3555 RTAC | SEL-3350 RTAC

Siemens DIGSI 5 | SIPROTEC 5 7SJ85 | SIPROTEC 5 7UM85

POWER & SUBSTATION · DATA & OPERATIONS

IEC 62351 (security for 61850 / DNP3 / 104) — data model & operations

62351 is a set of security parts, not a data protocol. The 'objects' are the mechanisms each part adds; enumeration is confirming which part is actually implemented and on.

FUNCTIONS & SERVICES

Operation	Code / where	Access
TLS handshake	62351-3	R
DNP3-SA challenge	62351-5	R
Signed GOOSE	62351-6	R
RBAC role check	62351-8	R

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

62351-3 (TLS) — TLS for the TCP profiles - MMS (3782), DNP3, IEC 104

62351-5 (auth) — Challenge-response authentication for DNP3/101/104 - the basis of DNP3-SA

62351-6 (GOOSE/SV) — Authentication/signature for the L2 multicast protocols

62351-8 (RBAC) — Role-based access control + the PKI/key management to support it

HOW READ / WRITE / COMMAND / POLL WORKS

Wrap: 62351 layers onto the base protocol: a TLS tunnel, an HMAC challenge, or a signature - the data model is unchanged.

Authenticate: DNP3-SA challenges critical operations; MMS/104 run inside TLS with mutual certificates.

Govern: RBAC + a PKI (e.g. SICAM GridPass) decide who holds which role.

POWER & SUBSTATION · ON THE WIRE

IEC 62351 (security for 61850 / DNP3 / 104) — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
62351-3	TLS record	TLS on MMS 3782, DNP3, IEC 104 TCP
62351-5	g120 objects	DNP3-SA challenge-response (HMAC) of critical ops
62351-6	auth tag	Signature on GOOSE/SV frames
62351-8	role	RBAC + the PKI/key management behind it

EXAMPLE — IS THE SECURITY LAYER ACTUALLY ON?

-> TCP 3782 ClientHello -> TLS = IEC 61850 SecureTLS present

DNP3-SA: master op -> outstation CHALLENGE (g120 v1) -> master HMAC reply (g120 v2)

GOOSE: 62351-6 appends an authentication tag to the PDU

record which parts are enabled; cleartext links need compensating segmentation

NATIVE READ-ONLY ENUMERATION

Look for SecureTLS ports (MMS 3782) and TLS on DNP3/104 links

Detect a DNP3-SA challenge-response in capture

Read each device's security config - which parts are supported and enabled

Verify certificate management / PKI (GridPass-style)

Record cleartext links that rely on compensating segmentation

POWER & SUBSTATION

Metering: ION & DLMS/COSEM

The protocols of revenue and power-quality meters: Schneider's proprietary ION protocol and the international DLMS/COSEM standard, alongside Modbus and DNP3 on the same devices.

HOW IT WORKS

ION PROTOCOL

Schneider PowerLogic meters (ION9000/PM8000/ION7650) speak the ION protocol on TCP 7700 (and Secure ION/TLS 7443) - a register/framework model for measurements, setup and logs, configured via ION Setup with an encrypted .scf security file (up to 50 users in advanced mode).

DLMS/COSEM

Device Language Message Specification / COSEM object model (IEC 62056) is the global smart-metering standard: COSEM interface objects (OBIS codes) accessed with association-level security (low/high-level authentication, optional ciphering). Meters also expose Modbus, DNP3 and BACnet.

MITRE ATT&CK for ICS **T0861 Point & Tag Identification** · **T0868 Detect Operating Mode** · **T0842 Network Sniffing**

TRANSPORT & PORTS

ION protocol TCP 7700; Secure ION/TLS 7443

DLMS TCP 4059

Also Modbus 502, DNP3 20000, BACnet 47808

HTTPS/SFTP on newer meters

SECURITY POSTURE

- ION: advanced security mode adds users/roles + TLS
- Legacy cleartext services (HTTP/FTP/Telnet) must be off
- DLMS supports HLS authentication + ciphering
- Lost meter passwords are unrecoverable - custody matters

EXAMPLE PRODUCTS

PowerLogic ION9000 · Juniper EX3400 · Juniper SRX340 · PowerLogic PM8000 · Phoenix Contact FL NAT 2000 · PxC FL SWITCH 20xx/21xx

POWER & SUBSTATION · DATA & OPERATIONS

Metering: ION & DLMS/COSEM — data model & operations

ION meters expose a framework of measurement/setup registers on TCP 7700; DLMS/COSEM meters expose OBIS-coded objects read with a GET inside an authenticated association.

FUNCTIONS & SERVICES

Operation	Code / where	Access
ION read register	TCP 7700	R
DLMS GET	read	R
DLMS SET	write	W
DLMS ACTION	method	Cmd
Association (LLS/HLS)	auth	R
Modbus FC03 read	502	R

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

ION framework — Modules + registers for measurements, setup and logs; .scf security file

COSEM objects — Interface classes (Register, Clock, Association LN) addressed by OBIS codes

OBIS code — A.B.C.D.E.F identifier, e.g. 1.0.0.0.0.255 = device ID

Also Modbus / DNP3 / BACnet — The same meter often answers several protocols at once

HOW READ / WRITE / COMMAND / POLL WORKS

ION: ION Setup opens the meter on 7700 (secure 7443) and reads framework registers; advanced mode adds users/roles.

DLMS: The client opens an association (low- or high-level security), then GETs COSEM objects by OBIS code.

Multi: A revenue/PQ meter typically answers Modbus and DNP3 for SCADA alongside ION/DLMS.

POWER & SUBSTATION · ON THE WIRE

Metering: ION & DLMS/COSEM — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
ION protocol	TCP 7700	Schneider register/framework model (secure 7443)
DLMS APDU	n bytes	AARQ association, then GET/SET/ACTION
COSEM object	class + OBIS	e.g. Register class, OBIS 1.0.0.0.0.255
Security	-	DLMS LLS/HLS authentication + optional ciphering

EXAMPLE — DLMS ASSOCIATION + GET BY OBIS (TCP 4059)

-> AARQ (association, LLS/HLS auth) <- AARE accepted

-> GET class 1 (Register) OBIS 1.0.1.8.0.255 (active energy)

<- value + scaler/unit

ION meters: read framework registers on 7700; also Modbus FC03 / DNP3 Class 0

NATIVE READ-ONLY ENUMERATION

ION protocol read of framework registers on 7700 (read-only)

DLMS association + GET by OBIS (e.g. 1.0.0.0.0.255 device ID)

Modbus FC03 / DNP3 Class 0 on the same meter

Confirm advanced/secure mode and that legacy HTTP/FTP/Telnet are off

Check the self-signed cert was replaced and verify revenue/anti-tamper lock

BUILDING AUTOMATION

BACnet/IP & MS/TP

The dominant building-automation protocol (ASHRAE 135): an object-oriented model of BAS devices, carried over UDP/IP (BACnet/IP) or RS-485 (MS/TP).

HOW IT WORKS

OBJECT / SERVICE MODEL

Every device is a collection of standard objects (Analog Input, Binary Output, Schedule, Trend Log…) with properties, accessed by ReadProperty / WriteProperty and event/COV subscriptions. Devices are addressed by a Device Instance number plus network/MAC address.

BACNET/IP VS MS/TP

BACnet/IP uses UDP 47808 (0xBAC0) with BVLL; BBMDs relay broadcasts across IP subnets (Foreign Device registration). MS/TP is a token-passing master/slave bus over RS-485. Who-Is/I-Am broadcasts enumerate the internetwork.

MITRE ATT&CK for ICS **T0846 Remote System Discovery** · **T0855 Unauthorized Command Message** · **T0836 Modify Parameter**

TRANSPORT & PORTS

BACnet/IP UDP 47808 (0xBAC0)

MS/TP over RS-485 (token bus)

BBMD / Foreign Device for routing

BACnet/SC (TLS) is the secure variant

SECURITY POSTURE

- No authentication or encryption natively
- WriteProperty accepted without credentials
- Who-Is/I-Am reveal the whole internetwork
- Control = segmentation, restrict BBMD/FD, BACnet/SC

EXAMPLE PRODUCTS

Benshaw RSi H2 VFD

Danfoss VLT FC102

PowerLogic ION9000

Innotech OMWEB02

PowerLogic PM8000

PowerLogic PM5560

BUILDING AUTOMATION · DATA & OPERATIONS

BACnet/IP & MS/TP — data model & operations

Every BAS device is a set of standard objects with numbered properties. Who-Is finds devices; ReadProperty pulls any property; WriteProperty commands - all unauthenticated.

FUNCTIONS & SERVICES

Operation	Code / where	Access
Who-Is / I-Am	discovery	Disc
ReadProperty	service 12	R
ReadPropertyMultiple	service 14	R
WriteProperty	service 15	W
SubscribeCOV	service 5	Sub
AtomicReadFile	service 6	R

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

Objects — Device, Analog Input (AI), Binary Output (BO), Schedule, Trend Log, Notification Class

Properties (numbered) — Object_Name=77, Present_Value=85, Vendor_Name=121, Object_List=76

Device instance — The device's identity (0 - 4194302), plus network + MAC address

BBMD / Foreign Device — Relay broadcasts across IP subnets; FD registration joins remotely

HOW READ / WRITE / COMMAND / POLL WORKS

Discover: A Who-Is broadcast makes every device answer I-Am with its instance and address.

Read: ReadProperty(Device, Object_List) lists the objects; then read Present_Value (85) on each point.

Command / COV: WriteProperty sets a point; SubscribeCOV pushes change-of-value notifications to the client.

BUILDING AUTOMATION · ON THE WIRE

BACnet/IP & MS/TP — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
BVLC	4 bytes	0x81 type, function (0x0A unicast / 0x0B broadcast), length
NPDU	2+ bytes	Version + control (routing)
APDU type	1 byte	0x10 unconfirmed, 0x00 confirmed request
Service	1 byte	0x08 Who-Is, 0x0C ReadProperty, 0x0F WriteProperty
Object / property	n bytes	Object id + property id (85 Present_Value, 77 Object_Name)

EXAMPLE — WHO-IS BROADCAST, THEN READPROPERTY

-> 81 0B 00 0C 01 20 FF FF 00 FF 10 08

BVLC broadcast | NPDU | APDU unconfirmed 0x10 | service 0x08 Who-Is

<- I-Am: device instance 1001, vendor id, max APDU

-> ReadProperty device:1001 property 85 (Present_Value) <- 72.4

NATIVE READ-ONLY ENUMERATION

Who-Is (BVLC 0x81, function 0x0A, unconfirmed service 8) - lists devices

ReadProperty Device:0 - Object_Name(77), Vendor_Name(121), Object_List(76)

ReadProperty Present_Value(85) on points (read-only)

Enumerate BBMD/FDT tables read-only

WriteProperty (15) is control - never on live BAS points

BUILDING AUTOMATION

BACnet/SC (Secure Connect)

The security modernization of BACnet: a WebSocket/TLS hub-and-spoke datalink that replaces vulnerable UDP broadcasts with authenticated, encrypted connections.

HOW IT WORKS

HUB-AND-SPOKE

BACnet/SC nodes connect outbound over secure WebSockets (TLS) to a hub, which relays BACnet NPDUs between them - eliminating broadcast-based discovery and the BBMD problem. The same BACnet object/service model runs on top.

TRUST MODEL

Each node holds an operational certificate; the hub authenticates nodes by certificate. This gives BACnet real transport authentication, integrity and confidentiality - but only where devices and the head-end support it (newer Desigo/Trane/Schneider firmware).

MITRE ATT&CK for ICS **T0830 Adversary-in-the-Middle** · **T0846 Remote System Discovery** · **T0859 Valid Accounts**

TRANSPORT & PORTS

WebSocket over TLS (hub port)

Hub-and-spoke datalink

Operational + issuer certificates

Same BACnet objects/services above

SECURITY POSTURE

- TLS authentication, integrity, confidentiality
- Removes UDP broadcast / BBMD exposure
- Requires cert management + capable firmware
- Mixed estates still bridge to legacy BACnet/IP

EXAMPLE PRODUCTS

PowerLogic PM5560

Siemens Desigo CC

Siemens Desigo PXC

Trane Symbio 800

BUILDING AUTOMATION · DATA & OPERATIONS

BACnet/SC (Secure Connect) — data model & operations

The BACnet objects and services are unchanged - the difference is underneath: a TLS WebSocket to a hub replaces UDP broadcast, so devices are reached by certificate, not by Who-Is flooding.

FUNCTIONS & SERVICES

Operation	Code / where	Access
Connect to hub (TLS WS)	BVLC-SC	R
ReadProperty	service 12	R
ReadPropertyMultiple	service 14	R
WriteProperty	service 15	W
SubscribeCOV	service 5	Sub

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

Same BACnet objects — Device/AI/BO/Schedule... with the same numbered properties

Hub + nodes — Nodes connect outbound to a hub that relays NPDUs - no broadcast domain

Operational certificate — Each node's identity, validated by the hub

VMAC — Virtual MAC addressing over the secure datalink

HOW READ / WRITE / COMMAND / POLL WORKS

Connect: Each node opens an outbound secure WebSocket (TLS) to the hub and authenticates with its certificate.

Relay: The hub forwards BACnet NPDUs between nodes - the same read/write services ride the secure link.

Interop: A bridge may still connect legacy BACnet/IP segments - that bridge is the weak point to check.

BUILDING AUTOMATION · ON THE WIRE

BACnet/SC (Secure Connect) — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
TLS WebSocket	-	Node -> hub, outbound, certificate-authenticated
BVLC-SC	n bytes	Secure-connect encapsulation (VMAC addressing)
NPDU / APDU	n bytes	Same BACnet services inside the secure link
Operational cert	X.509	Issued per node; the hub validates it

EXAMPLE — SECURE CONNECT THEN NORMAL READPROPERTY

-> node opens WSS (TLS) to hub, presents operational certificate

<- hub validates cert, relays NPDUs (no UDP broadcast)

-> ReadProperty device:1001 property 85 over the secure channel <- value

check any bridge back to legacy BACnet/IP

NATIVE READ-ONLY ENUMERATION

Confirm devices/hub run BACnet/SC (TLS) rather than legacy /IP

Verify the certificate chain and that the hub authenticates nodes

Run the same ReadProperty enumeration over the authorized secure channel

Check any bridge back to unsecured BACnet/IP

Record devices whose firmware cannot support /SC (compensating)

BUILDING AUTOMATION

LonWorks / LonTalk

A veteran building- and industrial-control networking technology (ISO/IEC 14908): a peer-to-peer control network of nodes exchanging network variables over twisted pair or IP.

HOW IT WORKS

NETWORK VARIABLES

Each node (historically a Neuron chip) publishes/subscribes to standard network variable types (SNVTs) bound to other nodes' variables - a distributed, peer-to-peer model configured by a network management tool, not a central master.

MEDIA & ADDRESSING

LonTalk runs over TP/FT-10 twisted pair, power line, or IP (LonWorks/IP, tunnelling over UDP). Nodes have a unique 48-bit Neuron ID; domains/subnets/nodes structure addressing. Authentication is an optional shared-key challenge, rarely used.

MITRE ATT&CK for ICS **T0842 Network Sniffing** · **T0855 Unauthorized Command Message** · **T0836 Modify Parameter**

TRANSPORT & PORTS

LonTalk over TP/FT-10, power line, or IP

LonWorks/IP tunnelling (UDP)

48-bit Neuron ID addressing

Optional shared-key authentication

SECURITY POSTURE

- Optional challenge-response auth; usually off
- No confidentiality
- The network management tool holds the trust
- Control = segmentation + physical access

EXAMPLE PRODUCTS

Danfoss VLT FC102 (LonWorks option)

BUILDING AUTOMATION · DATA & OPERATIONS

LonWorks / LonTalk — data model & operations

Nodes are bound to each other by network variables (SNVTs); there is no central master to poll. Enumeration means reading node identities and their bindings with the management tool.

FUNCTIONS & SERVICES

Operation	Code / where	Access
NV update (bound)	implicit	Pub/Sub
Explicit message read	read	R
Query Neuron ID / status	Disc	Disc
Wink (identify)	-	Disc
Config write	W	W

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

SNVT network variables — Typed input/output variables bound node-to-node

Neuron ID (48-bit) — The unique hardware identity of each node

Domain / subnet / node — The logical addressing hierarchy

Config properties (SCPT) — Per-node configuration values

HOW READ / WRITE / COMMAND / POLL WORKS

Bind: A management tool binds output NVs to input NVs; nodes then exchange updates peer-to-peer on change.

Read: The tool reads node identities, status and config via explicit messages.

Media: The same traffic runs over FT-10 twisted pair, power line, or tunnelled over IP.

BUILDING AUTOMATION · ON THE WIRE

LonWorks / LonTalk — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
L2 framing	-	TP/FT-10, power line, or LonWorks/IP (UDP tunnel)
Domain/Subnet/Node	-	Logical address; 48-bit Neuron ID is the hardware identity
NV update	n bytes	SNVT value pushed between bound variables
Auth	-	Optional shared-key challenge (usually off)

EXAMPLE — NETWORK-VARIABLE BINDING + READ

output NV nvoTemp (node A) -> bound -> input NV nviTemp (node B)
on change, node A pushes the SNVT value to node B
mgmt tool -> query Neuron ID + self-doc (read-only) -> node inventory
mgmt tool -> list bindings -> the control relationships

NATIVE READ-ONLY ENUMERATION

Query Neuron IDs and self-documentation via the management tool (read-only)

List the NV bindings to map the control relationships
Capture LonWorks/IP tunnel traffic where used
Confirm whether optional authentication is enabled
No binding or config changes on a live network

⚠ LonWorks enumeration generally needs the vendor management tool; no broad open dissector.

BUILDING AUTOMATION

Metasys N2

Johnson Controls' legacy building-automation fieldbus: a simple master/slave protocol over RS-485 linking a supervisor to field controllers and third-party devices.

HOW IT WORKS

BUS & FRAMING

A supervisor (master) polls N2 devices by address over RS-485, fixed at 9600 baud, no parity. Messages are ASCII-hex framed with a start character and checksum; the device replies. A daisy-chain topology with A/B (and Y/Z jumpered) wiring.

OBJECT REGIONS

Data is organized into object types - Analog Input/Output, Binary Input/Output and internal float/integer/byte points - that the master reads, overrides and releases. Widely used by VFDs (Danfoss, Benshaw) and legacy JCI equipment as an N2 slave.

MITRE ATT&CK for ICS **T0855 Unauthorized Command Message** · **T0836 Modify Parameter** · **T0868 Detect Operating Mode**

TRANSPORT & PORTS

RS-485, 9600 baud, 8-N-1

Master/slave polling by address

ASCII-hex framing + checksum

AI/AO/BI/BO + internal points

SECURITY POSTURE

- No authentication or encryption
- Address is identity; physical bus access = control
- Override/release commands accepted without auth
- Control = segmentation via gateway + physical access

EXAMPLE PRODUCTS

Benshaw RSi H2 VFD

Danfoss VLT FC102

BUILDING AUTOMATION · DATA & OPERATIONS

Metasys N2 — data model & operations

A supervisor polls each device address for its point values; points are organized into AI/AO/BI/BO regions plus internal types. The frame is ASCII-hex with a checksum.

FUNCTIONS & SERVICES

Operation	Code / where	Access
Read object / poll	poll	R
Read attribute	read	R
Override	write	W
Release	write	W

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

Device address — 1-255 on the RS-485 trunk - the device identity

AI / AO / BI / BO regions — Analog/binary input and output point regions

Internal ADF / ADI / BD — Internal float, integer and byte points

Frame — Start char + address + command + data + checksum (ASCII-hex)

HOW READ / WRITE / COMMAND / POLL WORKS

Poll: The supervisor addresses a device and reads a point region; the device returns the values.

Command: Override forces a point to a commanded value; Release returns it to local control.

Cycle: Polling repeats continuously - a simple master/slave loop at 9600 8-N-1.

BUILDING AUTOMATION · ON THE WIRE

Metasys N2 — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
Start char	1 byte	Frame start (ASCII)
Address	2 chars	Device address (ASCII-hex)
Command	2 chars	Read object / override / release
Data	n chars	Object type + point number + value (ASCII-hex)
Checksum / CR	3 chars	Checksum + carriage return

EXAMPLE — POLL AN ANALOG INPUT POINT (RS-485, 9600 8-N-1)

-> read AI point 1 on device 0x05 (ASCII-hex frame + checksum + CR)

<- device 0x05 returns the point value

regions: AI / AO / BI / BO + internal ADF/ADI/BD

override/release = writes; read-only poll only on live

NATIVE READ-ONLY ENUMERATION

Poll Read object / Read attribute from an authorized master (9600 8-N-1)

Map device addresses and the AI/AO/BI/BO regions present

Passive capture at an RS-485 tap decodes addr + command + checksum

Confirm any serial-to-IP gateway is restricted

Override / Release are writes - never on live BAS points

⚠ N2 ASCII-hex framing varies by device — confirm the exact start char and checksum against the JCI N2 spec.

BUILDING AUTOMATION

Siemens APOGEE P2 / FLN

The proprietary protocols of the Siemens APOGEE building-automation system: P2 (Protocol II) on the building-level network and FLN on the floor-level field bus.

HOW IT WORKS

P2 OVER ETHERNET

P2 field panels and the Insight supervisor exchange messages over TCP; each frame carries a length/sequence header and NUL-terminated ASCII routing slots naming the source and destination nodes within a building-level network (BLN). The BLN name is a cleartext admission label, not a secret.

FLN FIELD BUS

The Floor Level Network links terminal equipment controllers (TEC points) to the field panel over a serial bus. APOGEE also bridges to BACnet. Confirm exact framing against the site's version-matched Siemens documentation.

MITRE ATT&CK for ICS **T0842 Network Sniffing** · **T0846 Remote System Discovery** · **T0855 Unauthorized Command Message**

TRANSPORT & PORTS

P2 over Ethernet (TCP)

FLN serial floor-level bus

BACnet/IP bridge (47808)

BLN routing by ASCII node names

SECURITY POSTURE

- No authentication or encryption
- BLN name is a cleartext access label, not a secret
- Any node reaching the panel can exchange P2
- Control = segmentation + physical access

EXAMPLE PRODUCTS

Productivity2000 P2-550

Danfoss VLT FC102

Siemens APOGEE Insight

BUILDING AUTOMATION · DATA & OPERATIONS

Siemens APOGEE P2 / FLN — data model & operations

P2 moves point data between the Insight supervisor and field panels over TCP, routed by cleartext node names within a building-level network; FLN carries the floor-level TEC points.

FUNCTIONS & SERVICES

Operation	Code / where	Access
Read point value	P2	R
Command point	P2	Cmd
Route (BLN/node)	P2	Disc
BACnet bridge read	47808	R

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

P2 frame — Length/sequence header + NUL-terminated ASCII routing slots (src/dst node)

BLN — Building-Level Network name - a cleartext admission label in every frame

FLN TEC points — Terminal-equipment-controller points on the floor-level serial bus

Point database — Panel points read and commanded via Insight

HOW READ / WRITE / COMMAND / POLL WORKS

Exchange: The Insight supervisor and panels exchange P2 frames over TCP, routed by ASCII node names within the BLN.

Floor level: Panels poll TEC points over the FLN serial bus.

Bridge: APOGEE can expose points to BACnet/IP for integration.

BUILDING AUTOMATION · ON THE WIRE

Siemens APOGEE P2 / FLN — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
Frame header	-	Total length, header length, sequence number, direction
Routing slots	4 x ASCII	[BLN, dst-node, BLN, src-node] NUL-terminated
Opcode	2 bytes	AP2 function code (request/push frames)
Body	n bytes	Point data / command
BLN	ASCII name	Admission label in cleartext on every frame

EXAMPLE — P2 OVER TCP (BUILDING-LEVEL NETWORK)

-> P2 frame: len/seq header | [BLN,'PANEL2',BLN,'INSIGHT'] | opcode | body

BLN name is an access label, visible in cleartext (not a secret)

-> read point value via the Insight supervisor (authorized)

FLN carries floor-level TEC points on a serial bus

NATIVE READ-ONLY ENUMERATION

Passive P2 capture - map BLN and node names and the panels present

Read points via the Insight supervisor (authorized, read-only)

Inventory FLN / TEC points through the supervisor

Confirm the APOGEE-to-BACnet bridge exposure is restricted

Proprietary framing - confirm exact fields against Siemens documentation

⚠ P2/FLN framing is proprietary — confirm exact fields against the site's version-matched Siemens docs.

FIELD, PROCESS & MACHINE

HART & WirelessHART

The protocol that carries digital data over the legacy 4-20 mA instrument loop: HART superimposes FSK signalling on the analog current, and WirelessHART meshes it over radio.

HOW IT WORKS

FSK OVER 4-20 MA

HART (Highway Addressable Remote Transducer) modulates a Bell-202 FSK signal (1200/2200 Hz) on top of the 4-20 mA process current, so the analog measurement and digital config/diagnostics coexist on the same wires. A master (handheld or DCS) polls the field device by address.

WIRELESSHART

IEC 62591: a self-organizing 2.4 GHz TDMA mesh of field devices to a gateway, with AES-128 encryption and per-network join keys - genuinely secured, unlike wired HART. HART-IP tunnels HART over Ethernet/UDP to gateways.

MITRE ATT&CK for ICS **T0861 Point & Tag Identification** · **T0836 Modify Parameter** · **T0832 Manipulation of View**

TRANSPORT & PORTS

Wired HART: Bell-202 FSK on 4-20 mA loop

WirelessHART: 2.4 GHz TDMA mesh (IEC 62591)

HART-IP over UDP/TCP 5094

Multidrop addressing

SECURITY POSTURE

- Wired HART: no authentication/encryption on the loop
- WirelessHART: AES-128 + join keys (secured)
- Physical loop access is the wired control
- HART-IP exposure needs segmentation

EXAMPLE PRODUCTS

E+H Promag P300/P500 (HART option)

HART & WirelessHART — data model & operations

HART is a command set over the loop: universal commands read identity and the primary variable, common commands read dynamic variables and tags - all by command number.

FUNCTIONS & SERVICES

Operation	Code / where	Access
Read Unique ID	Cmd 0	R
Read Primary Variable	Cmd 1	R
Read Dynamic Variables	Cmd 3	R
Read Tag/Descriptor/Date	Cmd 13	R
Read Device Status	Cmd 48	R
Config / write commands	device-specific	W

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

Dynamic variables — PV / SV / TV / QV - the primary and secondary process values

Device identity — Manufacturer, device type, unique ID (from command 0)

Command classes — Universal (0-30), common-practice, device-specific

Poll address — Multidrop address; write-protect jumper on the instrument

HOW READ / WRITE / COMMAND / POLL WORKS

Poll: A master (handheld or DCS) sends a command to the device poll address; the device replies with the data.

Coexist: The FSK digital signal rides on the 4-20 mA loop without disturbing the analog measurement.

Wireless: WirelessHART meshes the same commands to a gateway with AES-128 and join keys.

FIELD, PROCESS & MACHINE · ON THE WIRE

HART & WirelessHART — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
Preamble	5-20 bytes	0xFF sync for the FSK modem
Start delimiter	1 byte	Frame type (short/long address)
Address	1 or 5 bytes	Polling address or unique device id
Command	1 byte	0 Read Unique ID, 1 Read PV, 13 Tag/Descriptor
Byte count + data	n bytes	Payload
Checksum	1 byte	XOR longitudinal parity

EXAMPLE — COMMAND 0 (READ UNIQUE ID) THEN COMMAND 1 (READ PV)

-> Cmd 0 (Read Unique ID) <- manufacturer, device type, unique id

-> Cmd 1 (Read Primary Variable) <- 72.4 degC (units + value)

-> Cmd 13 (Tag/Descriptor/Date) <- 'TT-101', descriptor, date

HART-IP tunnels the same commands over UDP/TCP 5094

NATIVE READ-ONLY ENUMERATION

HART-IP (5094): Command 0 (Read Unique ID), Command 1 (Read PV), Command 13 (Tag/Descriptor/Date) (Tag/Descriptor/Date)

Command 48 (Read Device Status) for health

Wired loop: read with a HART handheld/modem (read-only)

WirelessHART: verify join-key management + gateway security

Confirm the write-protect jumper is set; no config writes on a live loop

FIELD, PROCESS & MACHINE

CAN bus & SAE J1939

The in-machine network of engines, gensets and vehicles: CAN provides the robust differential bus, and SAE J1939 defines the higher-layer message set for heavy-duty powertrains.

HOW IT WORKS

CAN ARBITRATION

Controller Area Network is a multi-master differential-pair bus where each message has an identifier that also sets priority; lower IDs win arbitration. Built for noise immunity and determinism, not security - any node can transmit any ID.

J1939 & EXPANSION

SAE J1939 layers on CAN a 29-bit identifier scheme with PGNs (Parameter Group Numbers) and SPNs (Suspect Parameter Numbers) carrying engine/genset data to and from the ECU. Vendors (Basler, Cummins, Kohler, B&R) use CAN internally for expansion modules and the engine ECU link.

MITRE ATT&CK for ICS **T0842 Network Sniffing** · **T0855 Unauthorized Command Message** · **T0814 Denial of Service**

TRANSPORT & PORTS

CAN 2.0A/B differential bus

J1939: 29-bit ID, PGNs/SPNs

CANopen for automation devices

Engine ECU link (J1939) on gensets

SECURITY POSTURE

- No authentication or encryption
- Any node can transmit any identifier
- Priority is by ID, not identity
- Control = physical bus access; keep off IT networks

EXAMPLE PRODUCTS

B&R APC/PPC 3100

B&R X20 PLC

Basler DGC-2020HD / MTU MGC-3000

Cummins PowerCommand PCC3300

Kohler/SDMO APM802

CAN bus & SAE J1939 — data model & operations

CAN frames carry a priority-encoding identifier; J1939 packs a PGN into that 29-bit ID, and each PGN carries defined SPNs. CANopen devices expose an object dictionary via SDO.

FUNCTIONS & SERVICES

Operation	Code / where	Access
Broadcast PGN	periodic	Pub
Request PGN	PGN 59904	R
CANopen SDO upload	read	R
CANopen SDO download	write	W
NMT state change	command	Cmd

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

29-bit J1939 ID — Priority + PGN (Parameter Group Number) + source address

PGN / SPN — A PGN groups signals; each SPN is one defined signal (e.g. engine speed)

CANopen object dictionary — Index/subindex objects read/written via SDO

PDO / SDO — Process data (cyclic) vs. service data (acyclic config)

HOW READ / WRITE / COMMAND / POLL WORKS

Broadcast: ECUs transmit PGNs cyclically; lower IDs win arbitration and go first - no addressing handshake.

Request: A node sends Request PGN (59904) to pull a specific PGN on demand.

CANopen: An SDO upload reads an object-dictionary entry; PDOs carry the fast cyclic process data.

CAN bus & SAE J1939 — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
Priority	3 bits	Arbitration priority (lower ID wins)
PDU format / specific	16 bits	Together form the PGN (parameter group)
Source address	8 bits	Transmitting ECU
Data	8 bytes	The SPNs packed per the PGN definition

EXAMPLE — ELECTRONIC ENGINE CONTROLLER 1 (EEC1, PGN 61444)

29-bit ID 0x0CF00400 -> prio 3 | PGN 0xF004 (61444) | src 0x00

data 8 bytes -> SPN 190 Engine Speed = e.g. 1500 rpm (bytes 4-5, 0.125 rpm/bit)

-> Request PGN 0xEA00 (59904) to pull a specific PGN on demand

CANopen: SDO upload reads object-dictionary index (read-only)

NATIVE READ-ONLY ENUMERATION

Passive CAN capture - decode PGNs and SPNs to inventory signals

Request PGN (59904) to read a specific PGN (read-only)

CANopen SDO upload (read) of object-dictionary indices

Inventory nodes/ECUs and confirm the bus is physically contained

No frame injection on a live engine/genset bus

FIELD, PROCESS & MACHINE

Vendor genset serial (PCCNet, MDEC)

The proprietary serial control networks of engine-genset controllers: Cummins PCCNet and MTU MDEC/SmartConnect - alongside Modbus and J1939 on the same devices.

HOW IT WORKS

PCCNET & MDEC

Cummins PowerCommand controllers use PCCNet, an RS-485 network between Cummins genset devices (control, HMI, I/O), plus Modbus RTU on a shared RS-485 port and J1939 to the ECU. MTU engines use MDEC/SmartConnect serial links. These are cleartext, unauthenticated, and sparsely documented publicly.

SECURITY REALITY

No user accounts or encryption; control rests on HMI passwords (Operator/Service/Engineering levels), a Mode Change password, physical enclosure locking and serial isolation. Any IP path is a separate gateway on its own sheet. Treat exact framing as vendor-confidential - confirm against the service manual.

MITRE ATT&CK for ICS **T0855 Unauthorized Command Message** · **T0800 Activate Firmware Update Mode** · **T0816 Device Restart/Shutdown**

TRANSPORT & PORTS

PCCNet RS-485 (Cummins device net)

Modbus RTU on shared RS-485

J1939 CAN to the ECU

MTU MDEC/SmartConnect serial

SECURITY POSTURE

- No user accounts or encryption
- HMI + Mode Change passwords are the native control
- Published manual passwords = treat as publicly known
- Control = locked enclosure, serial isolation, gateway limits

EXAMPLE PRODUCTS

Cummins PowerCommand PCC3300

Basler / MTU MGC-3000

Kohler / SDMO APM802

Vendor genset serial (PCCNet, MDEC) — data model & operations

The readable surface is the Modbus register map and the HMI parameter tree; PCCNet and MDEC are vendor device-nets whose framing is not public, so enumerate via Modbus and the panel.

FUNCTIONS & SERVICES

Operation	Code / where	Access
Modbus FC03/04 read	502 / RTU	R
HMI view parameters	panel	R
J1939 PGN read	CAN	R
Modbus write setpoint	FC06/16	W
Mode change	HMI + password	Cmd

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

Modbus register map — Vendor map (e.g. Cummins A029X159) of genset status/measurements

HMI parameter tree — Menus + password levels (Operator / Service / Engineering)

PCCNet device net — Cummins RS-485 link between control/HMI/I-O - proprietary

J1939 engine data — PGNs/SPNs from the ECU over CAN

HOW READ / WRITE / COMMAND / POLL WORKS

Read: SCADA reads the Modbus register map; the HMI displays faults/history (viewing needs no password).

Device net: PCCNet links Cummins devices; MDEC/SmartConnect links MTU engines - cleartext serial.

Command: Mode changes and setpoints need the HMI/Mode-Change password levels.

FIELD, PROCESS & MACHINE · ON THE WIRE

Vendor genset serial (PCCNet, MDEC) — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
Modbus register map	-	Vendor map (e.g. Cummins A029X159) on RS-485/gateway
HMI tree	-	Menus + Operator/Service/Engineering password levels
PCCNet	RS-485	Cummins device net — proprietary framing
J1939	CAN	Engine ECU data (PGN/SPN)

EXAMPLE — READ GENSET STATUS VIA MODBUS (READ-ONLY)

-> 01 03 <start> <qty> <- status registers (voltage, freq, oil, faults)

viewing parameters/faults at the HMI needs no password

PCCNet / MDEC device-net framing is vendor-confidential

record any Modbus/IP gateway as a separate asset

NATIVE READ-ONLY ENUMERATION

Modbus FC03/FC04 reads of the published register map (read-only)

Read fault history / status at the HMI (viewing needs no password)

J1939 PGN capture for engine data

Record any Modbus/IP gateway as a separate asset

PCCNet / MDEC framing is vendor-confidential - confirm against the service manual; do not guess

⚠ PCCNet/MDEC are proprietary serial device-nets — enumerate via Modbus + the HMI and confirm the register map/framing against the service manual.

FIELD, PROCESS & MACHINE

Legacy RS-485: SY/MAX & monitor buses

The obsolete serial protocols still found on transformer monitors and temperature controllers: Square D SY/MAX and vendor Modbus-RTU links, reachable only through a gateway.

HOW IT WORKS

SY/MAX & MODBUS RTU

Square D's SY/MAX is a legacy master/slave serial protocol (selectable alongside Modbus RTU) on devices like the PowerLogic Model 98A temperature controller and Qualitrol 118 transformer monitor. A SCADA master reads setpoints and status over RS-485, typically bridged to IP by a serial-to-Ethernet / Modbus-TCP gateway.

PROTECTION INDEPENDENCE

These devices protect locally: over-temperature TRIP/FAN relays act on the thermocouple/RTD inputs independent of the comms link. That independence is the safety anchor - the network can be lost without losing protection.

MITRE ATT&CK for ICS **T0836 Modify Parameter** · **T0855 Unauthorized Command Message** · **T0880 Loss of Safety**

TRANSPORT & PORTS

RS-485 - SY/MAX or Modbus RTU

No native Ethernet; gateway to Modbus/TCP 502

Setpoint + status registers

Front-panel keypad + passcode (if any)

SECURITY POSTURE

- No accounts, no cryptography, no firmware update
- Unauthenticated serial; spoofable
- Cannot meet SL-3 natively - compensating controls
- Control = keypad lock, segmentation, lifecycle/replacement

EXAMPLE PRODUCTS

Square D Model 98A

Qualitrol 118 ITM

Legacy RS-485: SY/MAX & monitor buses — data model & operations

These monitors expose a small set of setpoint and status registers over Modbus RTU or SY/MAX, reached through a gateway; local protection runs independent of the link.

FUNCTIONS & SERVICES

Operation	Code / where	Access
Modbus FC03/04 read	RTU / gateway	R
SY/MAX read	serial	R
Write setpoint	FC06/16	W
Keypad read	panel	R

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

Setpoint registers — FAN / ALARM / TRIP thresholds + RS-485 address/baud

Status registers — Winding/ambient temperatures, relay states

Protocol select — Modbus RTU or SY/MAX (device-dependent)

Local protection — TRIP/FAN relays act on RTD/thermocouple inputs, not on comms

HOW READ / WRITE / COMMAND / POLL WORKS

Poll: A SCADA master reads setpoints/status over RS-485, usually via a serial-to-Modbus/TCP gateway.

Protect: The over-temperature TRIP/FAN logic runs locally on the sensor inputs - the network can fail without losing protection.

Legacy: No accounts or crypto; the device cannot meet SL-3 natively.

Legacy RS-485: SY/MAX & monitor buses — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
Protocol select	-	Modbus RTU or SY/MAX on RS-485
Registers	16-bit	Setpoint + status (temperature, relay state)
Gateway	-	Serial-to-Modbus/TCP 502 for SCADA
Local protection	-	TRIP/FAN relay acts on RTD/thermocouple, not comms

EXAMPLE — READ SETPOINTS/STATUS OVER THE GATEWAY (READ-ONLY)

-> 01 03 <start> <qty> (Modbus FC03 via the serial-to-IP gateway)

<- winding temperature, setpoints, relay states

confirm whether the link is Modbus RTU or SY/MAX

verify TRIP/FAN protection is independent of the comms link

NATIVE READ-ONLY ENUMERATION

Modbus FC03/FC04 reads via the gateway (502) from an authorized master

Confirm whether the link is Modbus RTU or SY/MAX

Read setpoints and status; verify local TRIP/FAN is independent of comms

Confirm the serial-to-IP gateway is restricted

Record as legacy/obsolete with a replacement plan

NETWORK INFRASTRUCTURE

Industrial wireless

The radio links that reach field devices and mobile assets: Wi-Fi, Bluetooth, cellular (LTE/5G), and the 802.15.4 mesh family (Zigbee / Thread / Trane Air-Fi).

HOW IT WORKS

WI-FI, BLUETOOTH, CELLULAR

Wi-Fi (802.11) links HMIs, laptops and bridges - secure with WPA2/3-Enterprise + 802.1X. Bluetooth appears on drives (ABB panel) and panel PCs for local config. Cellular (LTE/5G) provides remote links on gensets and edge PCs - treat any WAN path as untrusted, terminate in a firewall/VPN.

802.15.4 MESH

Zigbee / Thread (Schneider PAS600 sensor mesh) and Trane's BACnet/Zigbee Air-Fi (900 MHz) carry sensor/BAS data over low-power mesh with AES-128 network keys. Key management and pairing discipline are the controls; rogue joins are the risk.

MITRE ATT&CK for ICS **T0860 Wireless Compromise** · **T0830 Adversary-in-the-Middle** · **T0822 External Remote Services**

TRANSPORT & PORTS

Wi-Fi 802.11 (WPA2/3 + 802.1X)

Bluetooth (local config on drives/PCs)

Cellular LTE/5G (remote links)

802.15.4: Zigbee / Thread / Air-Fi (AES-128)

SECURITY POSTURE

- Wi-Fi security depends on WPA mode + 802.1X
- Bluetooth/local radios = physical-proximity exposure
- Cellular = untrusted WAN; VPN/firewall terminate it
- Mesh: AES-128 keys + controlled joining

EXAMPLE PRODUCTS

- ABB drive (Bluetooth)
- Schneider PAS600 (Zigbee/Thread)
- Trane Symbio 800 (Air-Fi)
- Cummins / Kohler (cellular)

Industrial wireless — data model & operations

Each radio has an identity and a security mode to read: an SSID and WPA/EAP config, a PAN ID + AES key, a Bluetooth device/pairing, a cellular APN. Enumeration is passive inventory - no active RF attacks.

FUNCTIONS & SERVICES

Operation	Code / where	Access
Wi-Fi associate / EAP	802.11	R
Mesh join (keyed)	802.15.4	R
BT pair / bond	BT	R
Cellular attach	LTE/5G	R

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

Wi-Fi — SSID / BSSID, WPA2/3 mode, 802.1X EAP method

802.15.4 mesh — PAN ID, channel, AES-128 network key (Zigbee / Thread / Air-Fi)

Bluetooth — Device class, pairing/bonding state

Cellular — APN, PLMN, SIM - a WAN path into the asset

HOW READ / WRITE / COMMAND / POLL WORKS

Associate: A client joins Wi-Fi via WPA/EAP, or a node joins the mesh using the network key, or a device bonds over Bluetooth.

Carry: Mesh and Wi-Fi carry sensor/BAS/HMI data; cellular backhauls to a remote endpoint.

Trust: AES keys and join/pairing discipline are the controls; a rogue join is the risk.

NETWORK INFRASTRUCTURE · ON THE WIRE

Industrial wireless — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
802.11 Wi-Fi	-	SSID/BSSID, WPA2/3 mode, 802.1X EAP
802.15.4 MHR	-	Frame control, seq, PAN id + addresses, aux security header
AES-128	-	Network key (Zigbee/Thread/Air-Fi); CCM* integrity+encrypt
Cellular	-	APN/PLMN; a WAN path to terminate in a firewall/VPN

EXAMPLE — PASSIVE INVENTORY ONLY (NO ACTIVE RF)

Wi-Fi survey -> SSID 'PLANT-OT' | BSSID | WPA2-Enterprise + 802.1X
802.15.4 -> PAN id 0x1A2B | channel 15 | security enabled (AES-128)
Bluetooth inquiry -> device list (drive panel, HMI)
no deauth / join / active RF tests; inventory cellular routers + where they land

NATIVE READ-ONLY ENUMERATION

Passive survey only: SSID/BSSID + security mode; PAN ID + channel; BT device inventory

Inventory cellular routers/gateways and where they terminate
Confirm WPA2/3-Enterprise + 802.1X on Wi-Fi
Verify cellular terminates in a firewall/VPN, not flat into OT
Check mesh key management + closed joining; no deauth / join / active RF tests

NETWORK INFRASTRUCTURE

Media redundancy (DLR, MRP, PRP/HSR, RSTP)

The Layer-2 protocols that keep OT networks up through a cable or switch failure - ring and parallel-path schemes tuned for the sub-second recovery control traffic demands.

HOW IT WORKS

RING PROTOCOLS

RSTP/MSTP is the generic spanning-tree; OT rings converge faster: DLR (Device Level Ring, EtherNet/IP), MRP (Media Redundancy Protocol, PROFINET) and vendor rings (HiPER-Ring, REP, Turbo Ring, N-Ring). A ring supervisor/manager blocks one path and heals on break.

PARALLEL REDUNDANCY

PRP (Parallel Redundancy Protocol) and HSR (High-availability Seamless Redundancy), IEC 62439-3, send every frame over two independent paths simultaneously for zero-loss failover - used in substations (SIPROTEC, EP6400) where a lost frame is unacceptable.

MITRE ATT&CK for ICS **T0814 Denial of Service** · **T0830 Adversary-in-the-Middle** · **T0842 Network Sniffing**

TRANSPORT & PORTS

RSTP/MSTP (spanning tree)

DLR (EtherNet/IP), MRP (PROFINET)

PRP/HSR (IEC 62439-3, seamless)

Vendor rings: HiPER-Ring, REP, Turbo/N-Ring

SECURITY POSTURE

- Control-plane protocols, not authenticated
- A rogue bridge/BPDU can disrupt topology
- Ring manager/supervisor role is sensitive
- Control = BPDU guard, root guard, managed switches

EXAMPLE PRODUCTS

Hirschmann RSP25 (HiOS) · Cisco Catalyst 9300 · Cisco IE3300 · Cisco IE4010 · Juniper EX3400 · Moxa EDS/MDS

NETWORK INFRASTRUCTURE · DATA & OPERATIONS

Media redundancy (DLR, MRP, PRP/HSR, RSTP) — data model & operations

These are control-plane protocols you read from the managed switches, not device data: ring roles and blocked ports (RSTP/DLR/MRP) or duplicate paths (PRP/HSR).

FUNCTIONS & SERVICES

Operation	Code / where	Access
BPDU exchange	RSTP	Disc
DLR beacon / announce	ring	Disc
MRP test frame	ring	Disc
PRP duplicate	parallel	Pub
HSR ring forward	ring	Pub

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

RSTP/MSTP — Bridge IDs, path cost, port roles (root/designated/blocked)

DLR / MRP — Ring supervisor/manager + ring ports; beacon/test frames

PRP — Redundancy Control Trailer; frame duplicated across LAN A/B

HSR — HSR tag; frame forwarded both ways around the ring

HOW READ / WRITE / COMMAND / POLL WORKS

Ring: A manager blocks one ring path to prevent a loop and unblocks it in milliseconds on a break.

Parallel: PRP/HSR send every frame over two paths at once - the receiver takes the first and drops the duplicate.

Read: The managed switches hold the live role/port state you enumerate.

NETWORK INFRASTRUCTURE · ON THE WIRE

Media redundancy (DLR, MRP, PRP/HSR, RSTP) — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
RSTP BPDU	-	Root id, bridge id, path cost, port role/flags
DLR / MRP	-	Ring supervisor/manager + beacon/test frames
PRP RCT	6 bytes	Trailer: seq number, LAN id, size, suffix 0x88FB
HSR tag	6 bytes	Path + size + seq, prepended to the frame

EXAMPLE — READ RING STATE FROM THE MANAGED SWITCHES

show spanning-tree -> root bridge, port roles (one blocked)
show dlr ring / show mrp -> supervisor/manager + ring ports
PRP: every frame duplicated on LAN A and LAN B (RCT suffix 0x88FB)
confirm BPDU guard / root guard on edge ports; no topology changes on live

NATIVE READ-ONLY ENUMERATION

show spanning-tree / show dlr ring / show mrp / show prp on the managed switches

Passive capture of BPDU / MRP / DLR to see the ring
Map ring roles, blocked ports and PRP/HSR LAN pairing
Confirm BPDU guard / root guard on edge ports
No topology-manipulation tests on live rings

NETWORK INFRASTRUCTURE

Time synchronization (NTP, PTP, IRIG-B)

The protocols that give every device a common clock - essential for sequence-of-events, protection coordination, TLS validity and forensic correlation across the plant.

HOW IT WORKS

NTP & PTP

NTP/SNTP (UDP 123) gives millisecond-level sync for logs and general use. PTP (IEEE 1588, the Power Profile in substations) gives sub-microsecond hardware-timestamped sync needed for Sampled Values and precise SOE - a grandmaster distributes time to boundary/transparent clocks.

IRIG-B

A dedicated time-code signal (often over a coax/fiber distribution) delivering precise time to relays and RTACs independent of the data network. Substation IEDs (SEL, SIPROTEC) commonly accept IRIG-B and/or PTP.

MITRE ATT&CK for ICS **T0830 Adversary-in-the-Middle** · **T0856 Spoof Reporting Message** · **T0814 Denial of Service**

TRANSPORT & PORTS

NTP/SNTP UDP 123

PTP (IEEE 1588) - L2/UDP, power profile

IRIG-B dedicated time code

Grandmaster / boundary clocks

SECURITY POSTURE

- NTP largely unauthenticated (symmetric keys optional)
- PTP spoofing skews SOE, protection and TLS validity
- Time source is a single point of trust
- Control = trusted internal time source, PTP profiles, monitoring

EXAMPLE PRODUCTS

Hirschmann RSP25 (HiOS) · CompactLogix 5380 · ControlLogix 5580 · ControlLogix 5590 · Innotech OMWEB02 · PxC FL SWITCH EP6400

NETWORK INFRASTRUCTURE · DATA & OPERATIONS

Time synchronization (NTP, PTP, IRIG-B) — data model & operations

Time is distributed by NTP (ms), PTP (sub-us, message-based), or IRIG-B (a continuous code). The enumeration is: what is each device's source, and is it trusted?

FUNCTIONS & SERVICES

Operation	Code / where	Access
NTP client poll	mode 3	R
PTP Sync / Follow_Up	grandmaster	Pub
PTP Delay_Req / Resp	slave<->master	R
Announce (BMCA)	election	Disc

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

NTP packet — Stratum, reference id, the four timestamps

PTP messages — Sync, Follow_Up, Delay_Req/Resp, Announce; domainNumber, clockClass

Grandmaster — The PTP time source; BMCA elects it

IRIG-B — A continuous BCD time-of-year code over dedicated wiring

HOW READ / WRITE / COMMAND / POLL WORKS

NTP: A client polls the server and disciplines its clock to within milliseconds.

PTP: The grandmaster multicasts Sync + Follow_Up; each slave measures path delay for sub-microsecond accuracy.

IRIG-B: Relays and RTACs read the continuous time code independent of the data network.

NETWORK INFRASTRUCTURE · ON THE WIRE

Time synchronization (NTP, PTP, IRIG-B) — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
NTP first byte	1 byte	LI (2b) + VN (3b) + Mode (3b), e.g. 0x1B = v3 client
Stratum / poll / prec	3 bytes	Clock quality indicators
Root delay / disp	8 bytes	Distance to the reference clock
Reference id	4 bytes	Upstream source
Timestamps	32 bytes	Reference, originate, receive, transmit

EXAMPLE — NTP CLIENT QUERY + PTP SYNC (READ-ONLY)

-> NTP mode 3 (0x1B ...) <- stratum 2, refid, server timestamps

PTP: grandmaster -> Sync + Follow_Up (multicast)

slave -> Delay_Req grandmaster -> Delay_Resp (sub-microsecond)

confirm each device's source is internal/trusted; no time injection

NATIVE READ-ONLY ENUMERATION

NTP mode-3 query (read-only) for stratum and reference id

Passive PTP capture - identify grandmaster, domain and offset

Confirm each device's time source and that it is internal/trusted

Check PTP domain/profile consistency on the process bus

No time-injection tests on live systems

NETWORK INFRASTRUCTURE

Discovery & provisioning (LLDP, HiDiscovery, BOOTP)

The Layer-2 protocols devices use to be found and addressed on commissioning - convenient for engineers, and a reconnaissance and tampering surface if left enabled.

HOW IT WORKS

TOPOLOGY & VENDOR DISCOVERY

LLDP (and CDP) advertise device identity/port/neighbor data for topology mapping. Vendor tools add their own: Hirschmann HiDiscovery (Layer-2, can set IP by MAC), ProSoft Discovery Service (PDS), Red Lion N-View, EtherNet/IP ListIdentity - all unauthenticated Layer-2/UDP.

ADDRESS PROVISIONING

BOOTP/DHCP (UDP 67/68) and rotary switches assign IP at install. Many adapters ship with a discovery/provisioning service and an embedded web server that must be disabled after commissioning (e.g. AENTR web auth-bypass, CISA ICSA-21-063-01).

MITRE ATT&CK for ICS **T0842 Network Sniffing** · **T0846 Remote System Discovery** · **T0873 Project File Infection**

TRANSPORT & PORTS

LLDP / CDP (L2 topology)

HiDiscovery (L2, set IP by MAC)

BOOTP/DHCP UDP 67/68

PDS, N-View, ListIdentity

SECURITY POSTURE

- Unauthenticated L2 discovery reveals inventory
- HiDiscovery read-write can change device IP
- Commissioning web servers often left enabled
- Control = disable/read-only after commissioning

EXAMPLE PRODUCTS

B&R X20 Modbus bus controller

Hirschmann RSP25 (HiOS)

Cisco Catalyst 9300

Cisco IE3300

Cisco IE4010

E+H Promag P300/P500

Discovery & provisioning (LLDP, HiDiscovery, BOOTP) — data model & operations

These protocols exist to announce or assign identity. Read passively they give a free topology and inventory; left in read-write they let an attacker re-address devices.

FUNCTIONS & SERVICES

Operation	Code / where	Access
LLDP advertise	periodic L2	Disc
CDP advertise	periodic L2	Disc
HiDiscovery read	L2	R
HiDiscovery set IP	L2	Cmd
BOOTP request	67/68	R
ListIdentity	0x63	Disc

R read · W write · Cmd command · Pub/Sub publish/subscribe · Disc discovery

DATA MODEL / OBJECTS

LLDP TLVs — Chassis ID, Port ID, System Name/Description, Management Address

HiDiscovery — Layer-2; can read device info and set IP by MAC

BOOTP/DHCP — UDP 67/68 address assignment at install

ListIdentity / PDS / N-View — Vendor discovery of EtherNet/IP, ProSoft, Red Lion devices

HOW READ / WRITE / COMMAND / POLL WORKS

Announce: Switches and devices multicast LLDP/CDP with their identity and neighbours - a ready-made topology map.

Provision: HiDiscovery/BOOTP assign IP at commissioning; many adapters also run a setup web server.

Harden: After commissioning these should be disabled or set read-only.

NETWORK INFRASTRUCTURE · ON THE WIRE

Discovery & provisioning (LLDP, HiDiscovery, BOOTP) — on the wire

How it actually looks: the frame layout, a real example exchange, and the native requests to enumerate it read-only.

FRAME / PDU ANATOMY

Field	Size	Meaning
Dst MAC	6 bytes	01:80:C2:00:00:0E (LLDP multicast)
EtherType	2 bytes	0x88CC (LLDP)
TLVs	n bytes	Chassis ID (1), Port ID (2), TTL (3), System Name/Desc...
End TLV	2 bytes	0x0000
Vendor discovery	-	HiDiscovery (L2), BOOTP (67/68), ListIdentity (0x63)

EXAMPLE — PASSIVE LLDP COLLECT + VENDOR DISCOVERY

<- 0x88CC | Chassis ID 'sw-core' | Port ID 'Gi1/0/3' | TTL 120 | Sys 'IE3300'

-> a free topology + management-IP map

HiDiscovery (read-only) -> device list by MAC; then set read-only/off

no HiDiscovery set-IP / BOOTP provisioning on live devices

NATIVE READ-ONLY ENUMERATION

Passively collect LLDP/CDP for topology + management IPs

HiDiscovery read-only to inventory (then set read-only/off)

SNMP LLDP-MIB read for neighbour tables

EtherNet/IP ListIdentity (0x63) for device nameplates

HiDiscovery set-IP / BOOTP provisioning - never issue on live devices

APPENDIX — GENERIC IT MANAGEMENT PROTOCOLS

Management protocols, not cyber-physical ones

Most products also expose these for management. They are in scope for hardening but are not control-layer protocols — verify encryption, disable the cleartext ones, and scope access.

Protocol	Port(s)	Role	Encrypted?	SL-3 note
SSH	TCP 22	Encrypted CLI management	Yes	Prefer over Telnet; disable if unused
HTTPS / TLS	TCP 443	Encrypted web UI / API	Yes	Replace self-signed certs; TLS 1.2+
SNMP	UDP 161/162	Monitoring / traps	v3 only	v3 auth+priv; read-only community
Syslog	514 / 6514 TLS	Event / audit logging	Over TLS	Ship to a central collector
RADIUS / TACACS+	1812/1813 · 49	Central AAA	Shared-secret	Per-device keys; TACACS+ cmd auth
Telnet / HTTP / FTP	23 · 80 · 21	Legacy cleartext mgmt	None	Disable everywhere at SL-3
RDP / SMB / WinRM	3389 · 445 · 5985/6	Windows host mgmt	Yes	NLA on; SMB signed, no v1; scope FW
NETCONF / RESTCONF	830 · 443	Network automation	Over SSH/TLS	Least-privilege login classes

PUTTING IT TO WORK

The read-only assessment workflow

Whatever the protocol, the method is the same: know the surface, enumerate without disturbing, judge against the requirement, and prove it with evidence.

1 · Identify

Confirm vendor, model, firmware and component type. List every protocol and port the device exposes — that is the attack surface.

2 · Enumerate read-only

Pull the native backup/config export, then use native read requests (identity reads, status polls, model browse) and passive capture. Never change keyswitch, download, force I/O or flash firmware on a live system.

3 · Assess vs. SL-3

For each protocol, check authentication, integrity and encryption against the 62443-3-3/4-2 requirement. Record Pass / Fail / Partial + compensating control + residual risk.

4 · Report

Executive summary, scope, top findings & mitigations, observations, technical appendix (evidence), references. Hash and store evidence on the encrypted share only.

SOURCES & STANDARDS

Where this comes from

Standards	ISA/IEC 62443-3-3 (system requirements) & 4-2 (component requirements); ISA-TR84.00.09; IEC 61850, IEC 60870-5, IEEE 1815 (DNP3), IEC 62351, IEC 62056 (DLMS), ASHRAE 135 (BACnet), IEC 62439-3 (PRP/HSR), IEEE 1588 (PTP).
Threat frameworks	MITRE ATT&CK for ICS (technique IDs + names shown throughout); MITRE EMB3D for device-level weaknesses.
Source material	The ISA/IEC 62443 SL-3 product cheat sheets — ports / interfaces and comms rows — plus vendor protocol manuals and CISA ICS advisories cited per device.
Caveats	Vendor-proprietary framing (PCCNet, MDEC, SY/MAX, ANSL/INA, P2/FLN) is documented sparsely — confirm exact details against the site's version-matched manual. Advisory status is time-sensitive; re-check at assessment time.